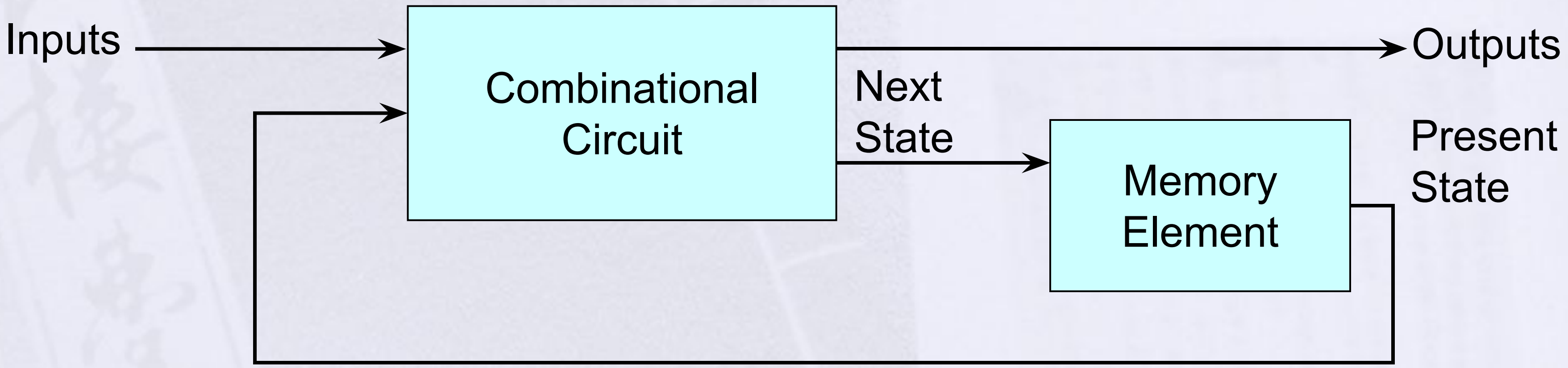
Synchronous Sequential Circuits

Sequential Circuits

- Sequential Circuits — consist of storage elements in addition to logic gates. Such circuits outputs are a function of the inputs and the state of the storage elements. The state of the storage element is a function of previous inputs.
- The outputs of sequential circuit depend on present input values and past inputs.

Sequential Circuit



Types of Sequential Circuits

Synchronous

• Behavior defined from knowledge of its signals at **discrete instances of time**.

• Storage elements observe inputs and can change state only in relation to a timing signal (clock pulses from a clock).

• **Asynchronous**

• Behavior defined from knowledge of inputs at **any instant of time** and the order in continuous time in which inputs change.

Memory/Storage Elements

- A storage element is a digital circuit that can maintain a binary state indefinitely
(as long as power is provided)

- Latches:
  - Storage elements that operate with signal levels are referred to as latches
  - SR and D are its types
  - A latch is a temporary storage device that has two stable states
- Flip Flops
  - Storage elements controlled by clock transition.

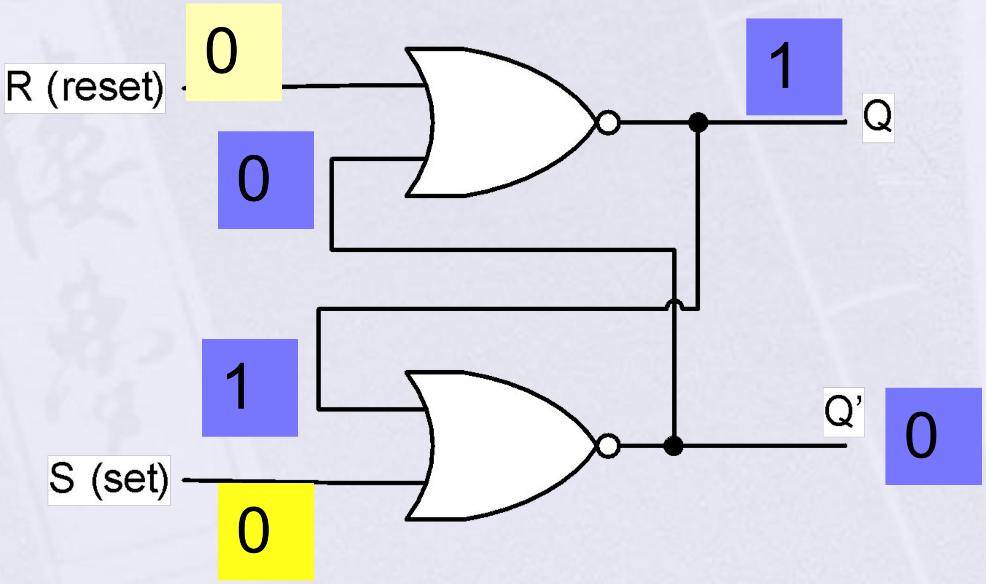
Latch v Flip-Flop

- The difference between a latch and a flip-flop is that a latch is asynchronous, and the outputs can change as soon as the inputs do (or at least after a small propagation delay).
- A flip-flop, on the other hand, is *edge-triggered* and only changes state when a control signal goes from high to low or low to high

SR Latch

- The S-R (Set-Reset) latch is the most basic type. It can be constructed from NOR gates or NAND gates.

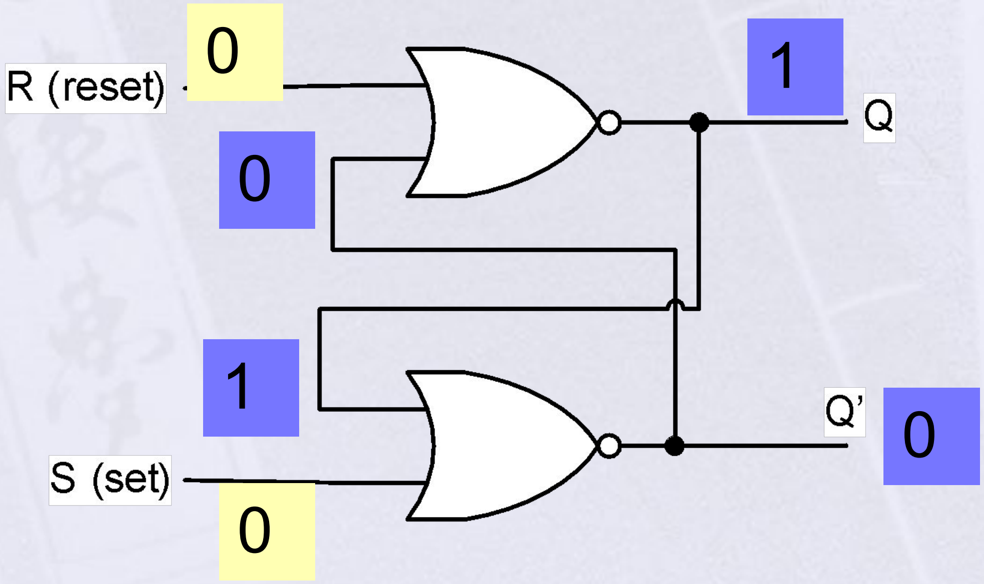
S-R Latch



S	R	Q	Q'
1	0	1	0
0	0	1	0
0	1	0	1
0	0	0	1
1	1	0	0

Function Table

S-R Latch

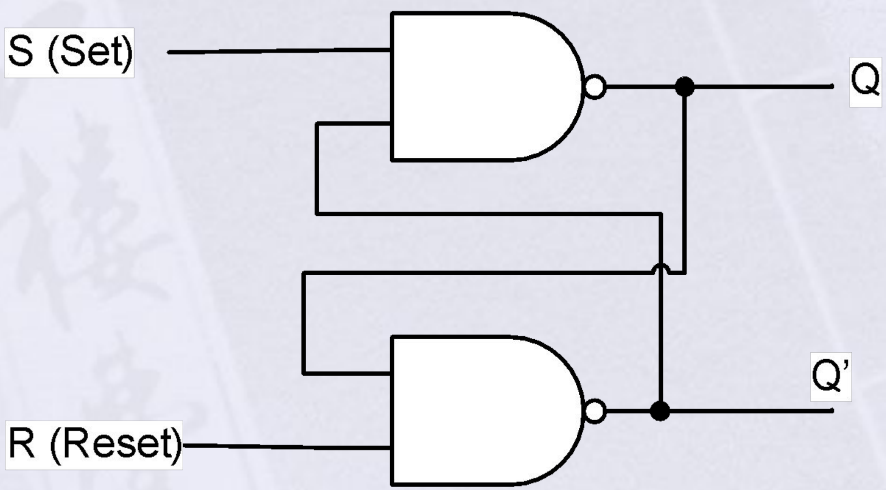


S	R	Q	Q'
1	0	1	0
0	0	1	0
0	1	0	1
0	0	0	1
1	1	0	0

Function Table

Indeterminate State

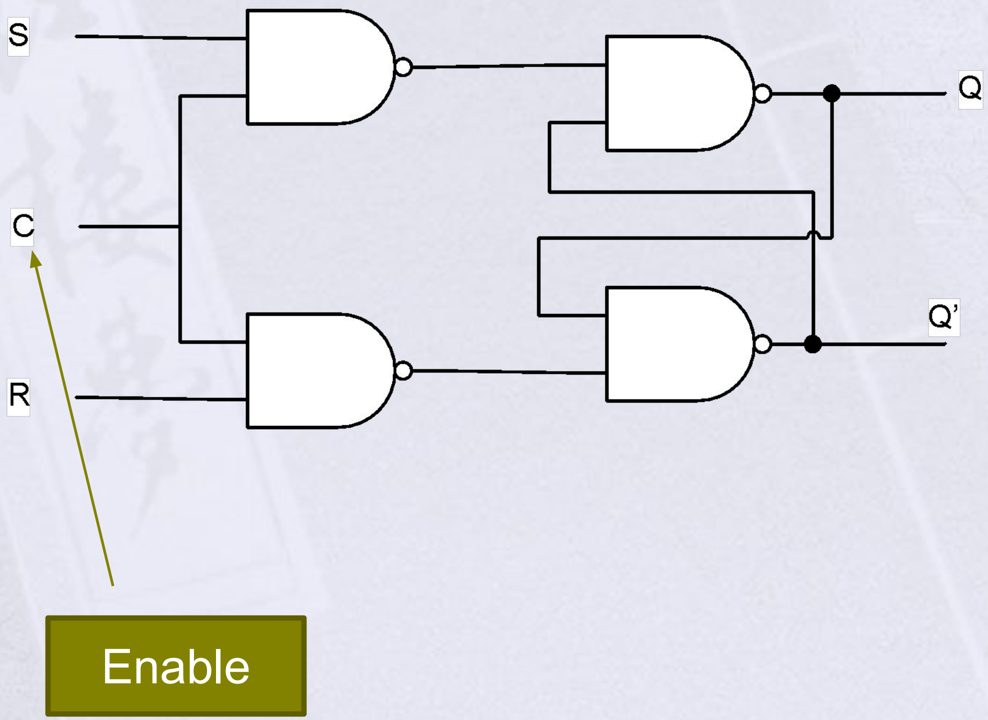
SR Latch using NAND Gate



S	R	Q	Q`
1	0	0	1
1	1	0	1
0	1	1	0
1	1	1	0
0	0	Indeterminate	

Function Table

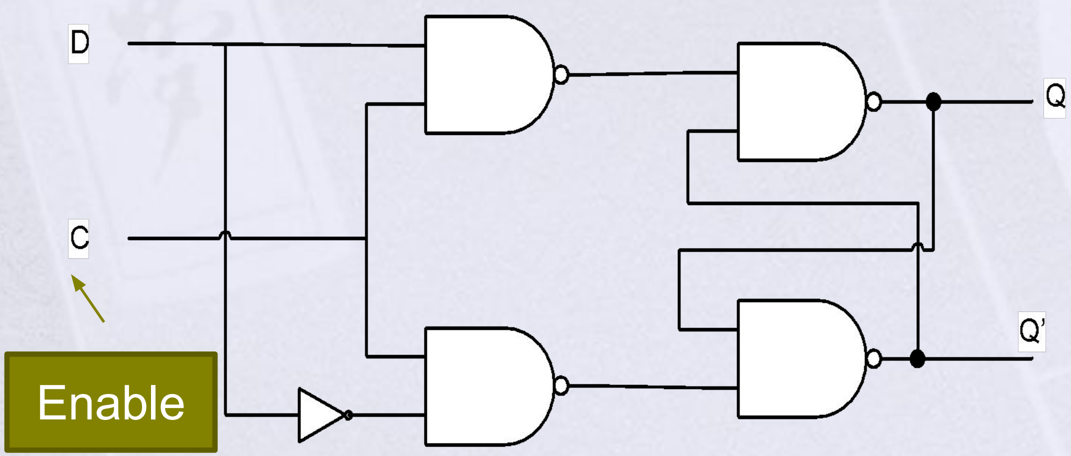
SR Latch using NAND Gate With Enable Input



C	S	R	Next state of Q
0	X	X	No Change
1	0	0	No Change
1	0	1	Q = 0; Reset State
1	1	0	Q = 1; Set State
1	1	1	Indeterminate

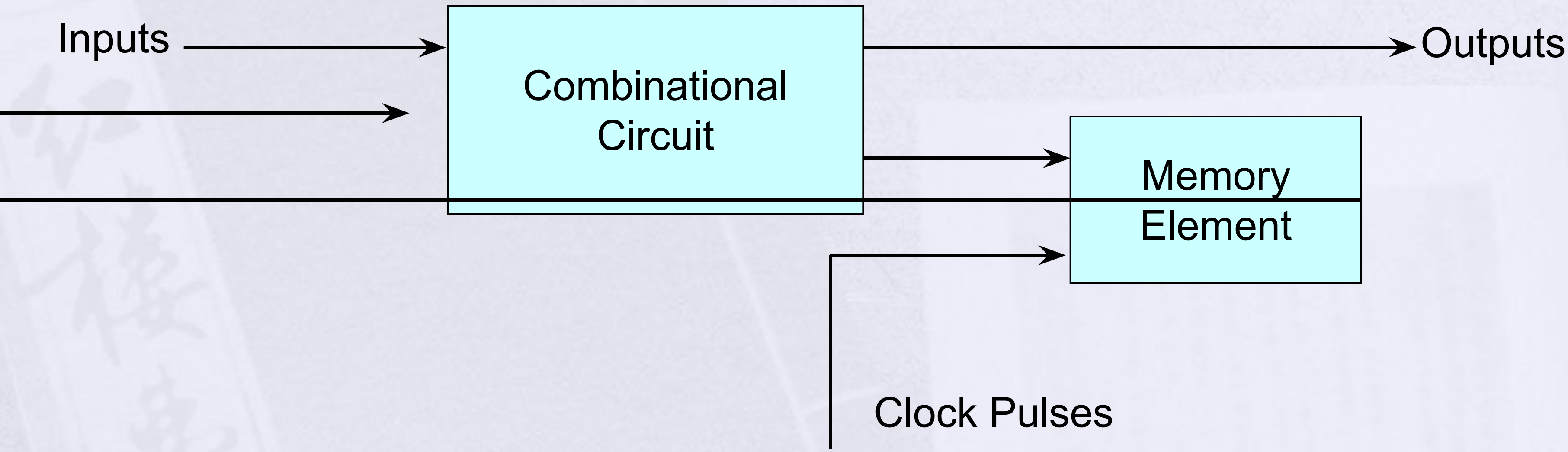
D-Latch

- The D latch is a variation of the S-R latch but combines the S and R inputs into a single D input
- Used as data storage device enabled by the clock
- Also called transparent latch



C	D	Next state of Q
0	X	No Change
1	0	Q=0; Reset State
1	1	Q=1; Set State

Synchronous Sequential Circuit



Timing Diagram of Clock Pulses

Flip-Flops

- The state of the flip-flop or a latch is switched by the change in the control input – C.
- This momentary change is called <u>trigger</u>

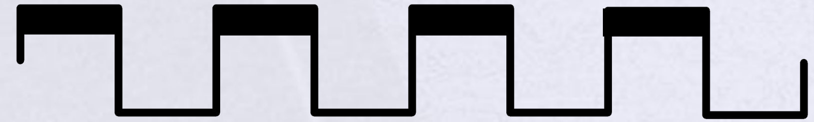
and the transition it causes is said to trigger the flip-flop.

- The D-Latch with pulses in its control input is essentially a flip-flop that is triggered every time the pulse goes to logic 1.

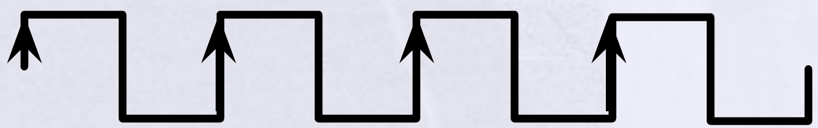
Flip-Flops

- Flip-flops are constructed in such a way to operate them properly when working in a proper clock.
- The key to the proper operation of flip-flop is to trigger it only during a *signal transition*.

Flip Flops



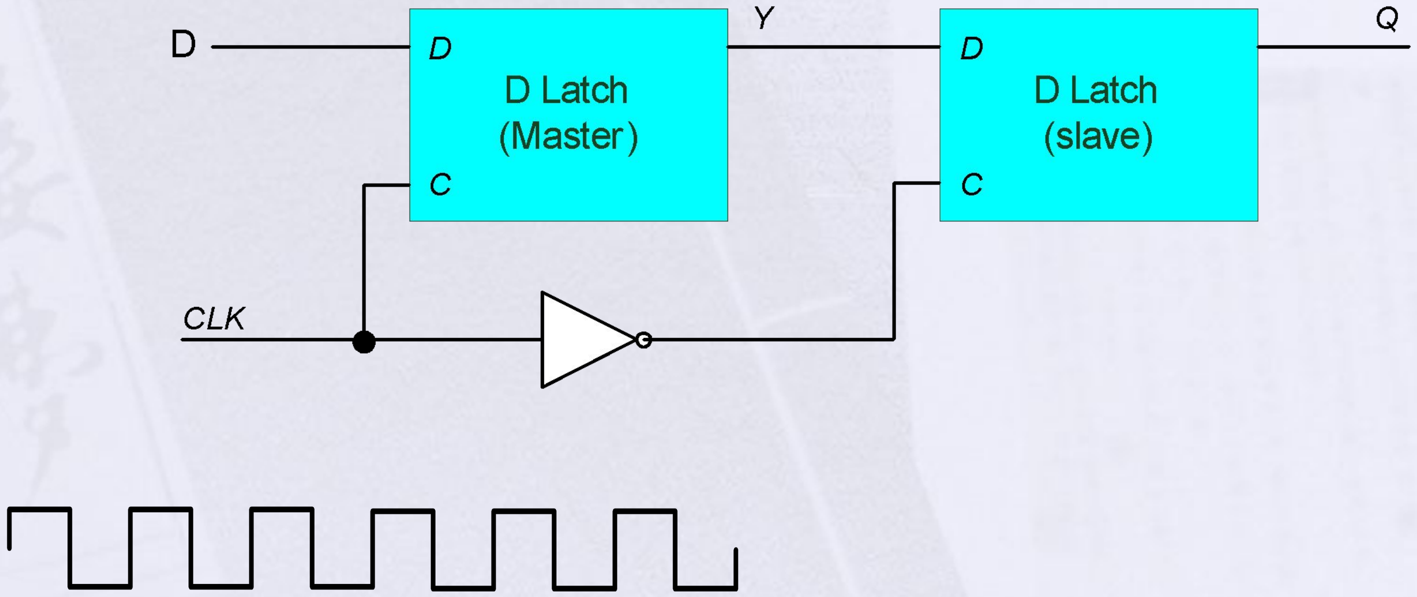
Response to Positive Level



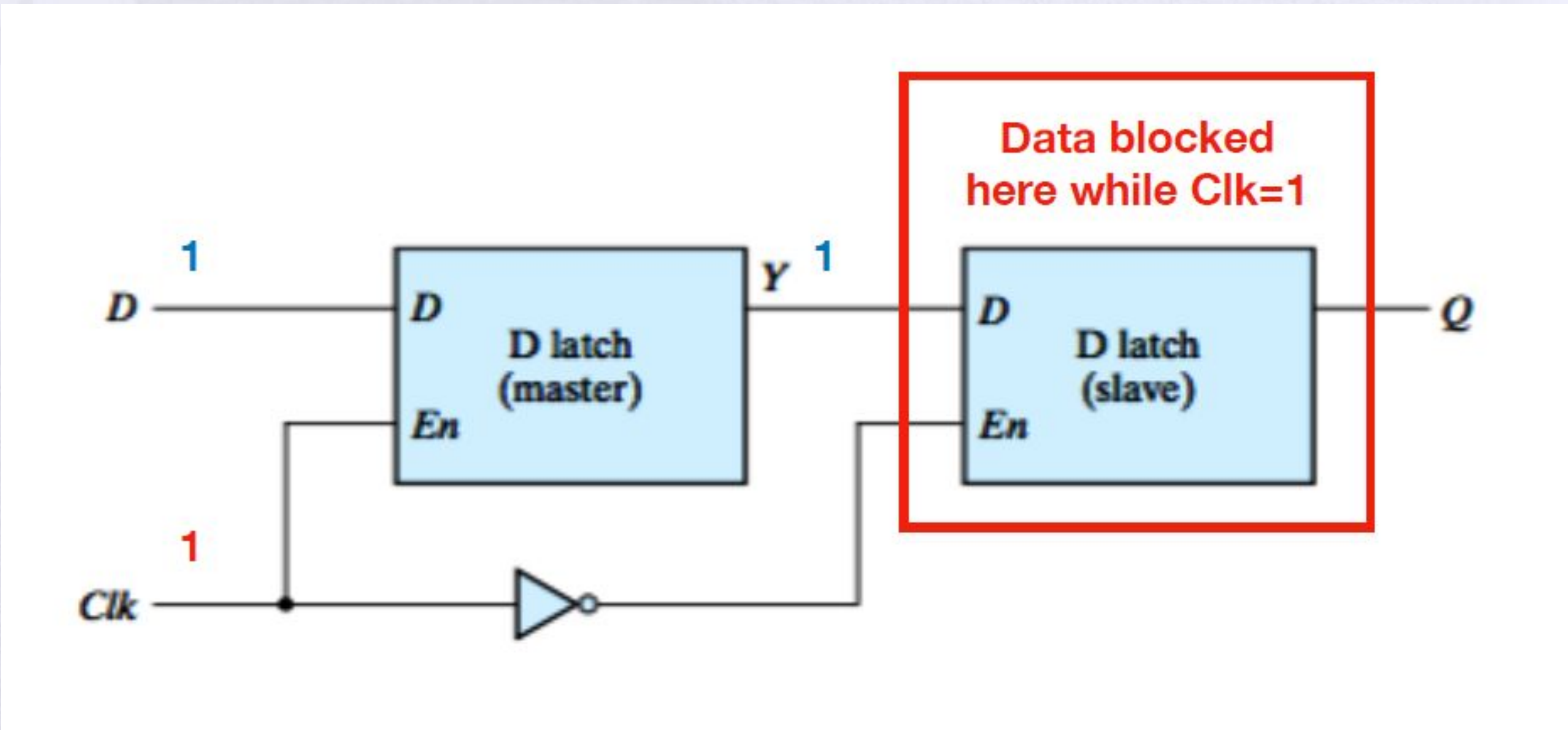
Positive-Edge Response

Negative-Edge Response

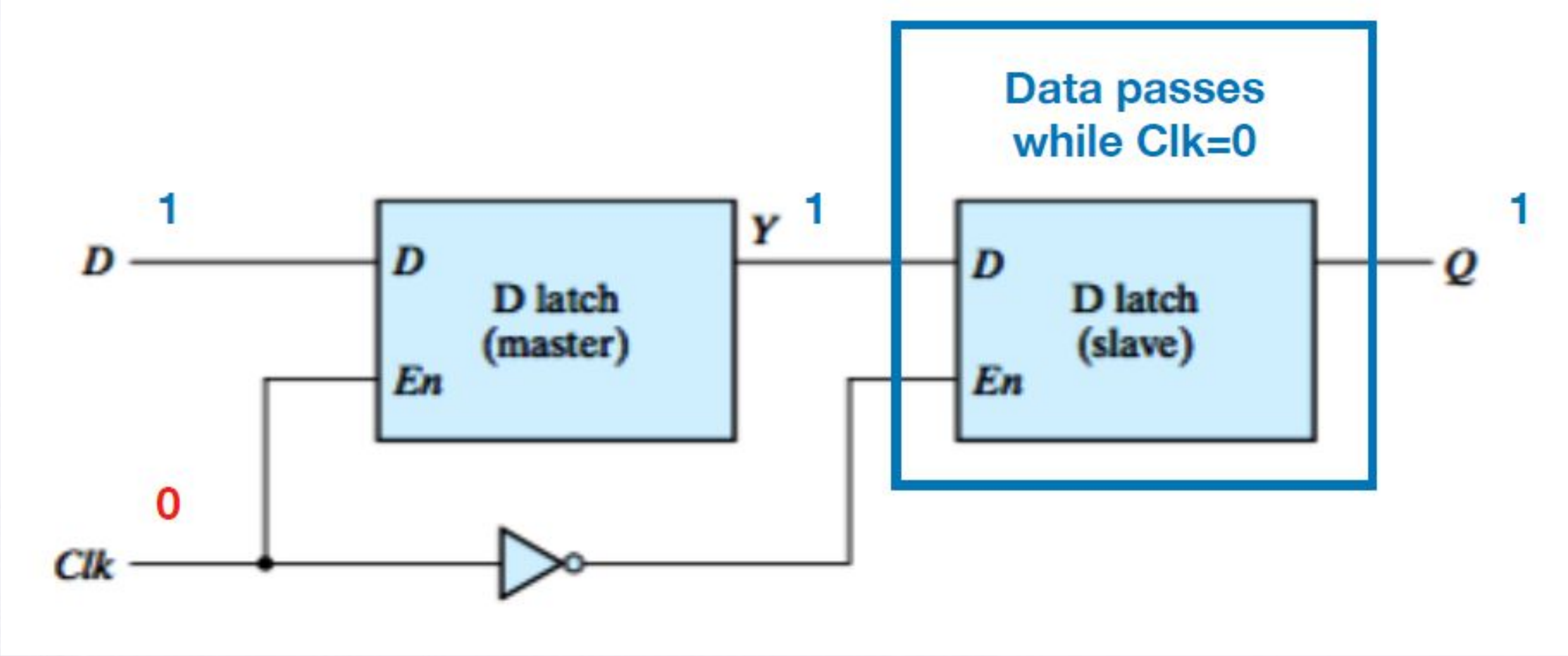
Edge-Triggered *D* Flip-Flop



Edge Triggered D Flip Flop

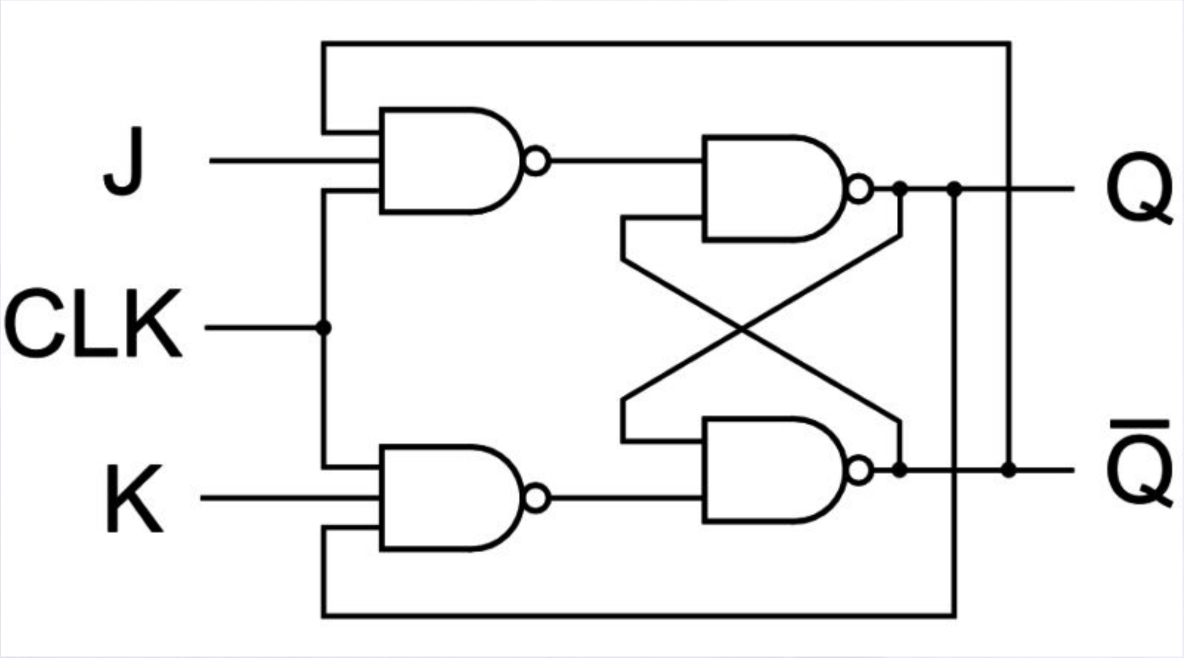


Edge Triggered D Flip Flop

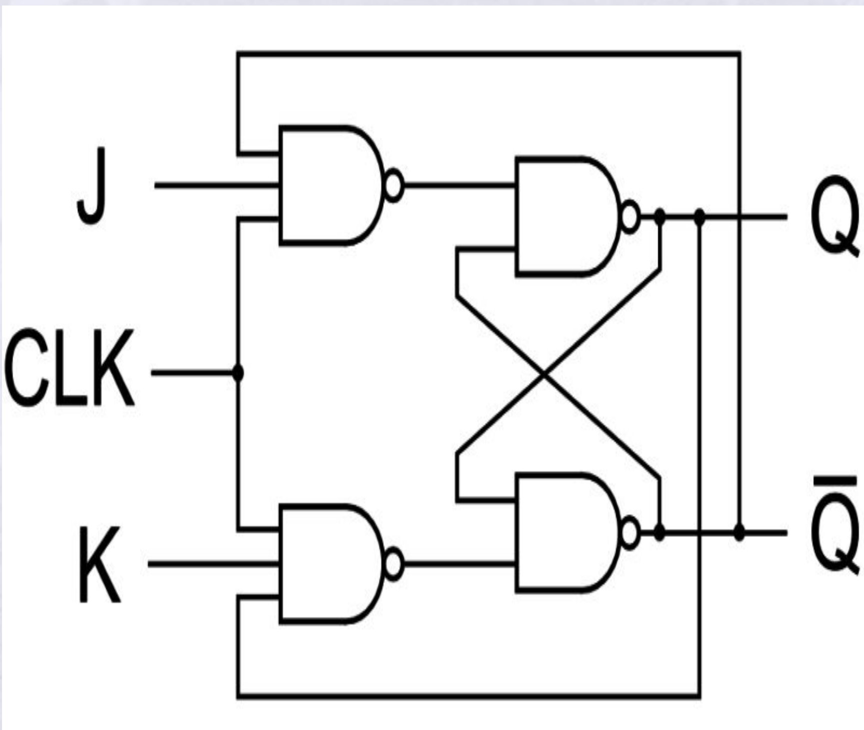


J-K Flip Flop

A J-K flip-flop is nothing more than **an S-R flip-flop with an added layer of feedback**. This feedback selectively enables one of the two set/reset inputs so that they cannot both carry an active signal to the multivibrator circuit, thus eliminating the invalid condition.

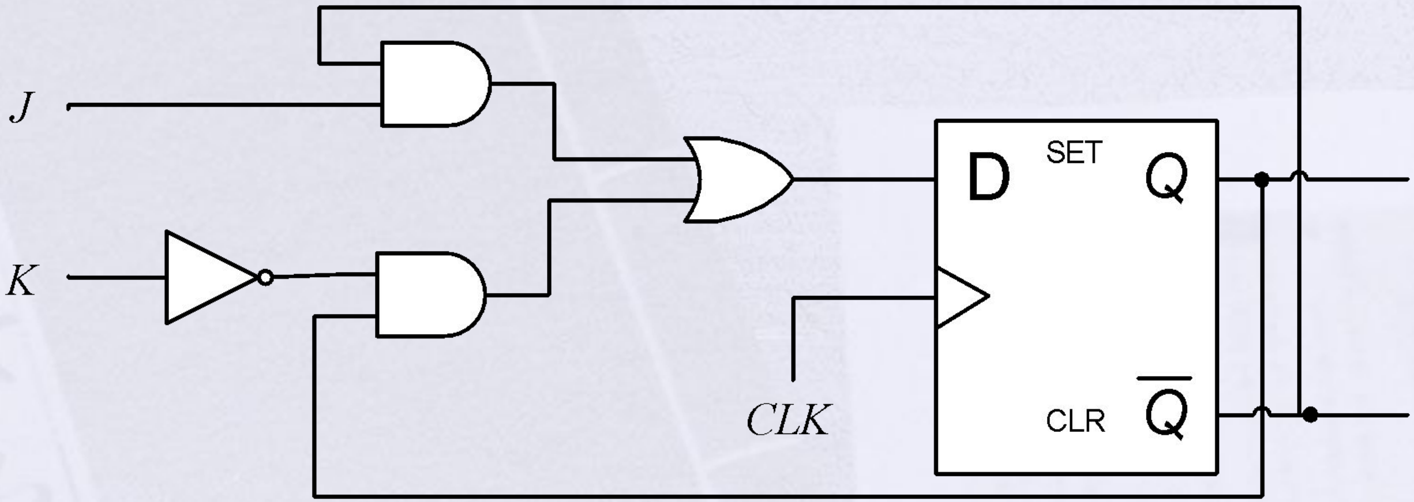


J-K Flip Flop

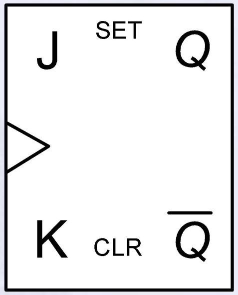


Trigger	Inputs		Output				Inference
			Present State		Next State		
CLK	J	K	Q	Q'	Q	Q'	
	x	x	-		-		Latched
	0	0	0	1	0	1	No Change
			1	0	1	0	
	0	1	0	1	0	1	Reset
			1	0	0	1	
	1	0	0	1	1	0	Set
			1	0	1	0	
	1	1	0	1	1	0	Toggles
			1	0	0	1	

J-K Flip Flop using D-FF



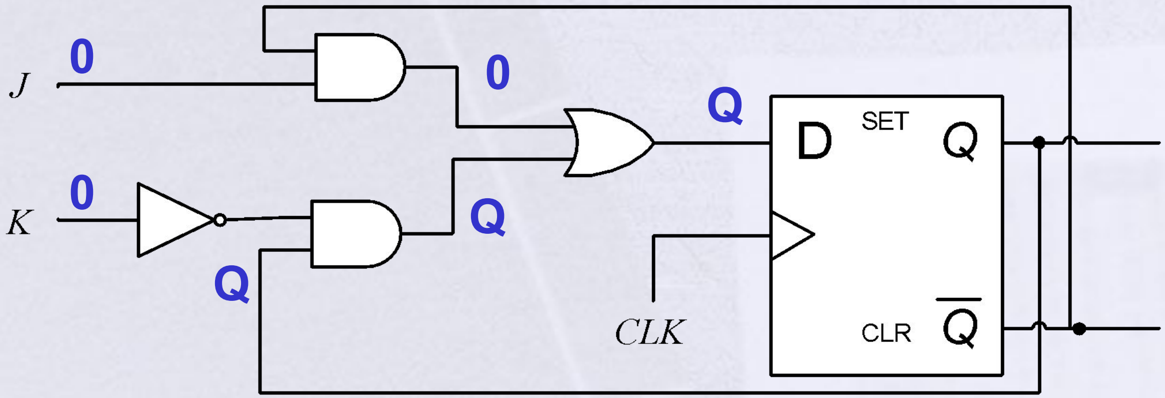
J	K	Q(t+1)	
0	0	Q(t)	No change
0	1	0	Reset
1	0	1	Set
1	1	Q'(t)	Complement



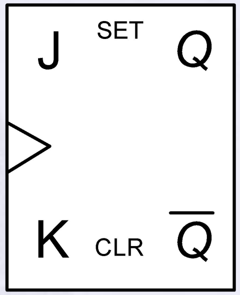
(Characteristic Equation)

Q(t+1) = JQ' + K'Q

J-K Flip Flop using D-FF



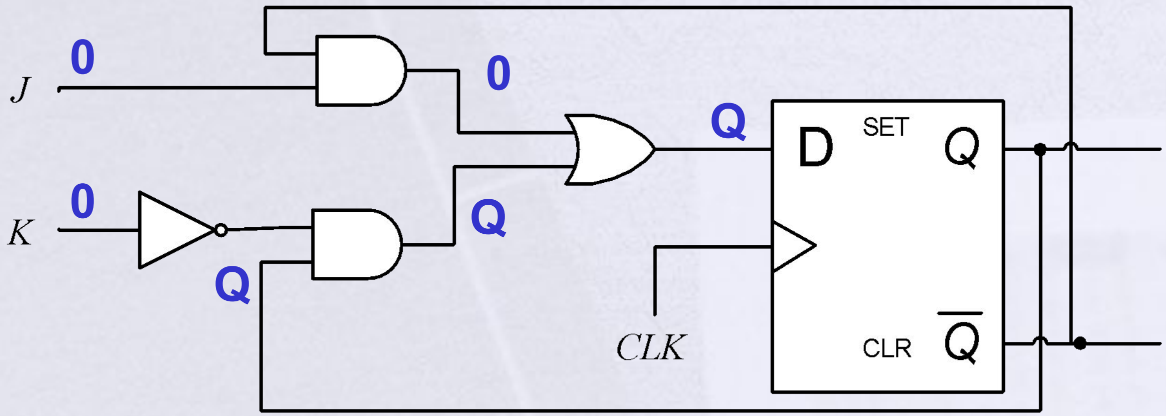
J	K	Q(t+1)	
0	0	Q(t)	No change
0	1	0	Reset
1	0	1	Set
1	1	Q'(t)	Complement



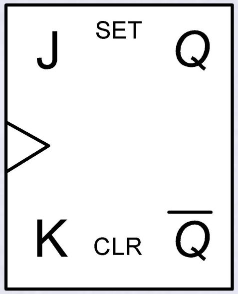
(Characteristic Equation)

Q(t+1) = JQ' + K'Q

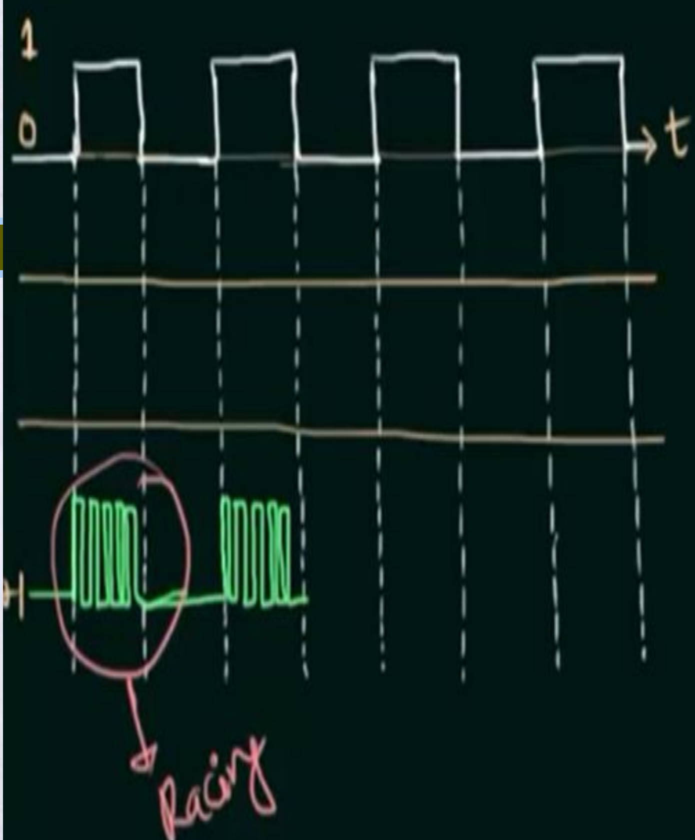
J-K Flip Flop

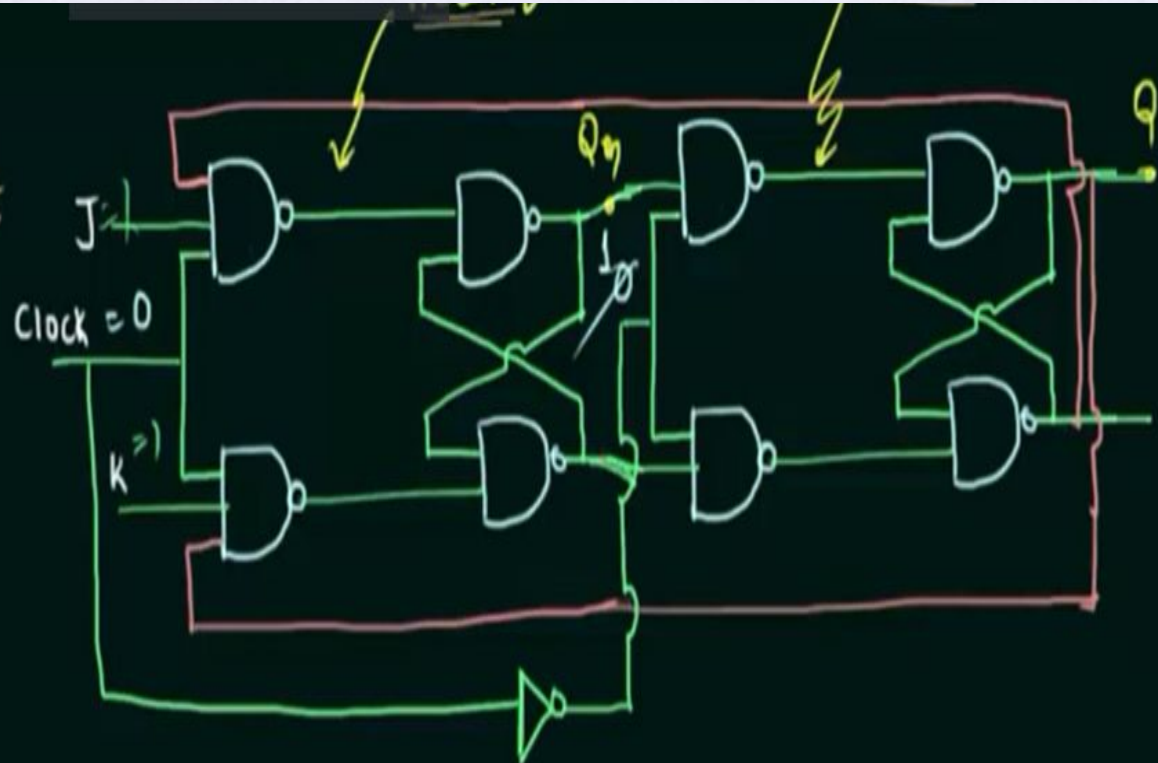


$Q(t+1) = JQ' + K'Q$ **(Characteristic Equation)**



J	K	Q(t+1)	
0	0	Q(t)	No change
0	1	0	Reset
1	0	1	Set
1	1	Q'(t)	Complement





i) T/2 < prop. delay

ii) Edge

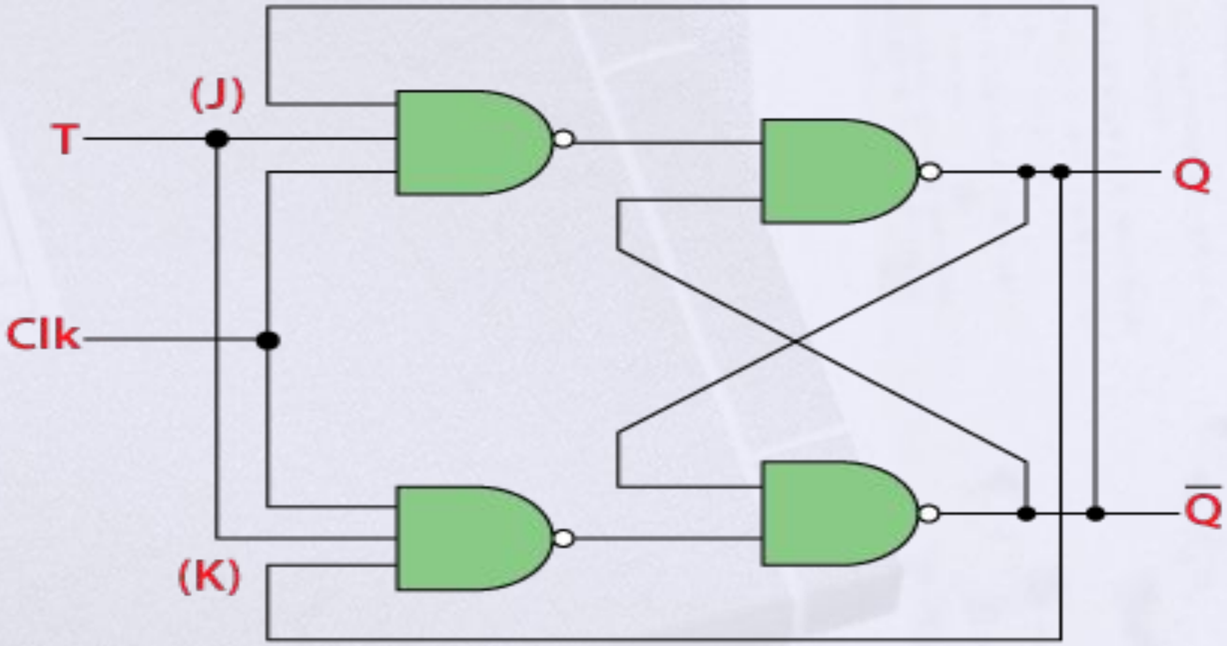
iii) MS op

* M.S. op. is same as ⊖ve edge trigg

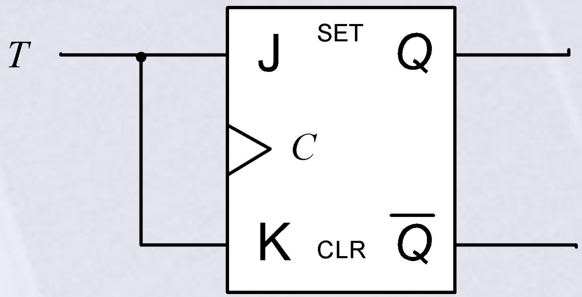
T-Flip Flop

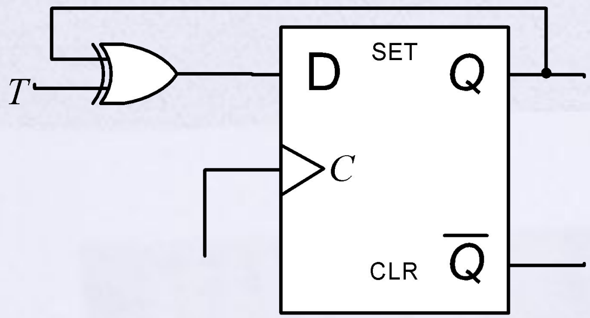
In D Flip-Flop: **When the clock rises from 0 to 1**, the value remembered by the flip-flop becomes the value of the D input (Data) at that instant.

In T Flip-Flop: When the clock rises from 0 to 1, the value remembered by the flip-flop either toggles or remains the same depending on whether the T input (Toggle) is 1 or 0.



T-Flip Flop



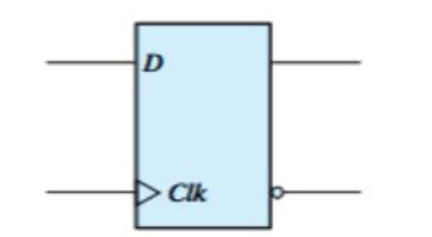


Q(t+1) = TQ' + T'Q **(Characteristic Equation)**

Input	Outputs	
	Present State	Next State
T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

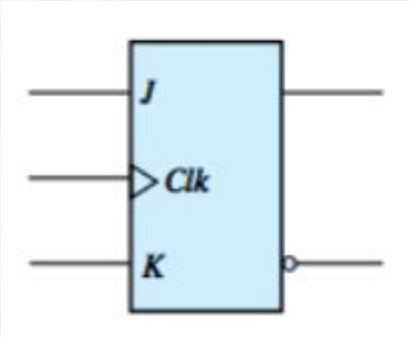
T	Q(t+1)
0	Q(t)
1	Q'(t)

Flip Flops – Characteristic Tables



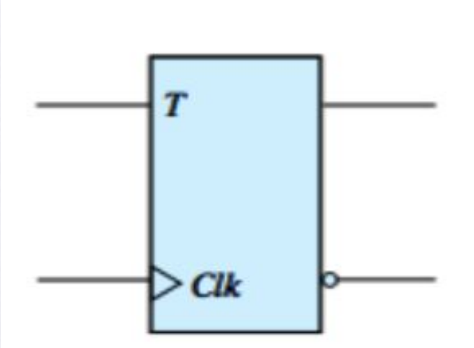
D Flip Flop

D	Q(t+1)
0	0 Reset
1	1 Set



J-K Flip Flop

J	K	Q(t+1)	
0	0	Q(t)	No change
0	1	0	Reset
1	0	1	Set
1	1	Q’(t)	Complement



T Flip Flop

T	Q(t+1)
0	Q(t)
1	Q’(t)

Flip Flops – Characteristic Equations

D Flip Flop

Q(t+1) = D

D	Q(t+1)
0	0 Reset
1	1 Set

J-K Flip Flop

Q(t+1) = JQ' + K'Q

J	K	Q(t+1)	
0	0	Q(t)	No change
0	1	0	Reset
1	0	1	Set
1	1	Q'(t)	Complement

T Flip Flop

Q(t+1) = TQ' + T'Q

T	Q(t+1)
0	Q(t)
1	Q'(t)

ANALYSIS OF CLOCKED SEQ CIRCUITS

Analysis of Clocked Seq Circuits

- **Analysis describes what a given circuit will do under certain operating conditions.**
- **The behaviour** of **a clocked sequential** circuit **is** determined from **the inputs, the outputs,** *and* **the state of** its **flip-flops.**

Analysis of Clocked Seq Circuits

1. State Equation (*aka transition equation*)
   - Specifies the next state as a function of present state and input
2. State Table (*aka transition table*)
   - Displays the relationship between various states in a table form
   - *Present State, Input, Next State, Output*
3. State Diagram
   - Graphic representation
   - States represented by circles
   - Lines represent transition

Analysis of Clocked Seq Circuits

State Table

- The state table: shows what the **next state** and the **output** will be as a function of the present state and the input:

Inputs of the combinational circuit		Outputs of the table	
Present State	*Input*	Next State	Output

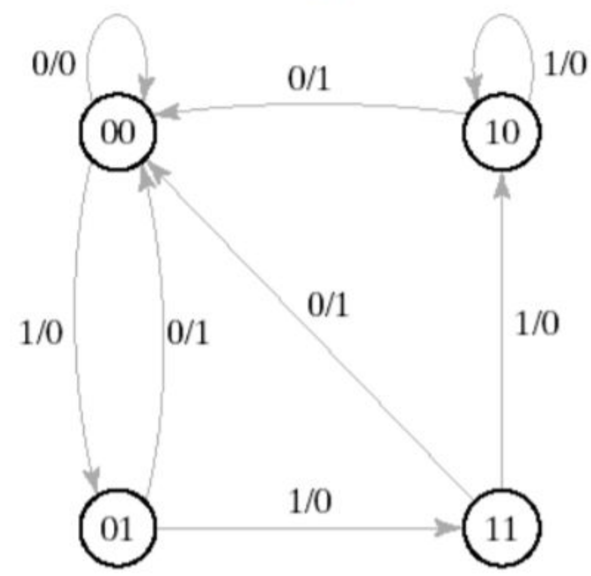
- The State Table can be considered a truth table defining the combinational circuits:
- the inputs are **Present State, Input**,
- and the outputs are **Next State and Output**.

Analysis of Clocked Seq Circuits

State Diagram:

The information available in a state table can be represented graphically in the form of a state diagram. In this type of diagram, a state is represented by a circle, and the transitions between states are indicated by directed lines connecting the circles.

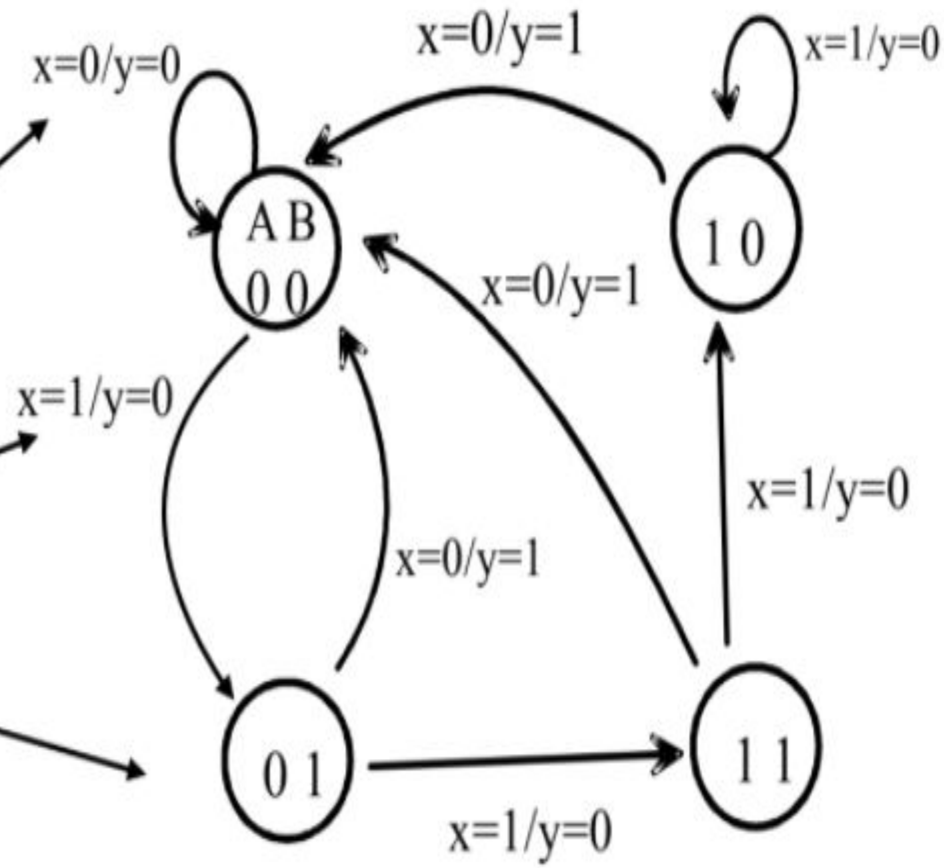
1/0 : means input =1
output=0



Analysis of Clocked Seq Circuits

Graphical representation of the State Table:

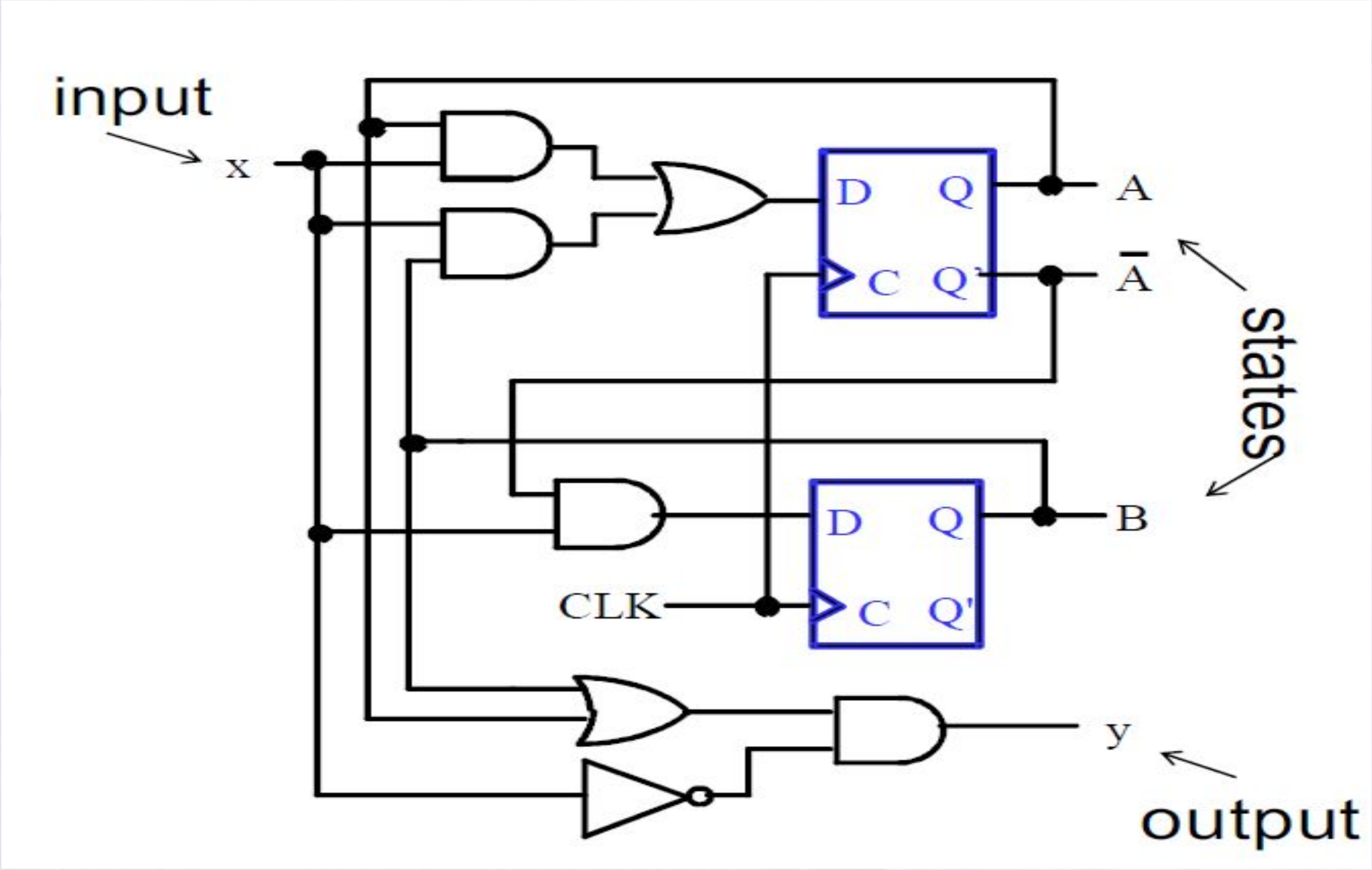
Present State		Input	Next State		Output
A(t)	B(t)	x(t)	A(t+1)	B(t+1)	y(t)
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0



Analysis of Sequential Circuit

- Find the input equations to the flipflops (next state equation) and the output equation
- Derive the state table
- Draw the state diagram

Analysis of Sequential Circuit



Analysis of Sequential Circuit

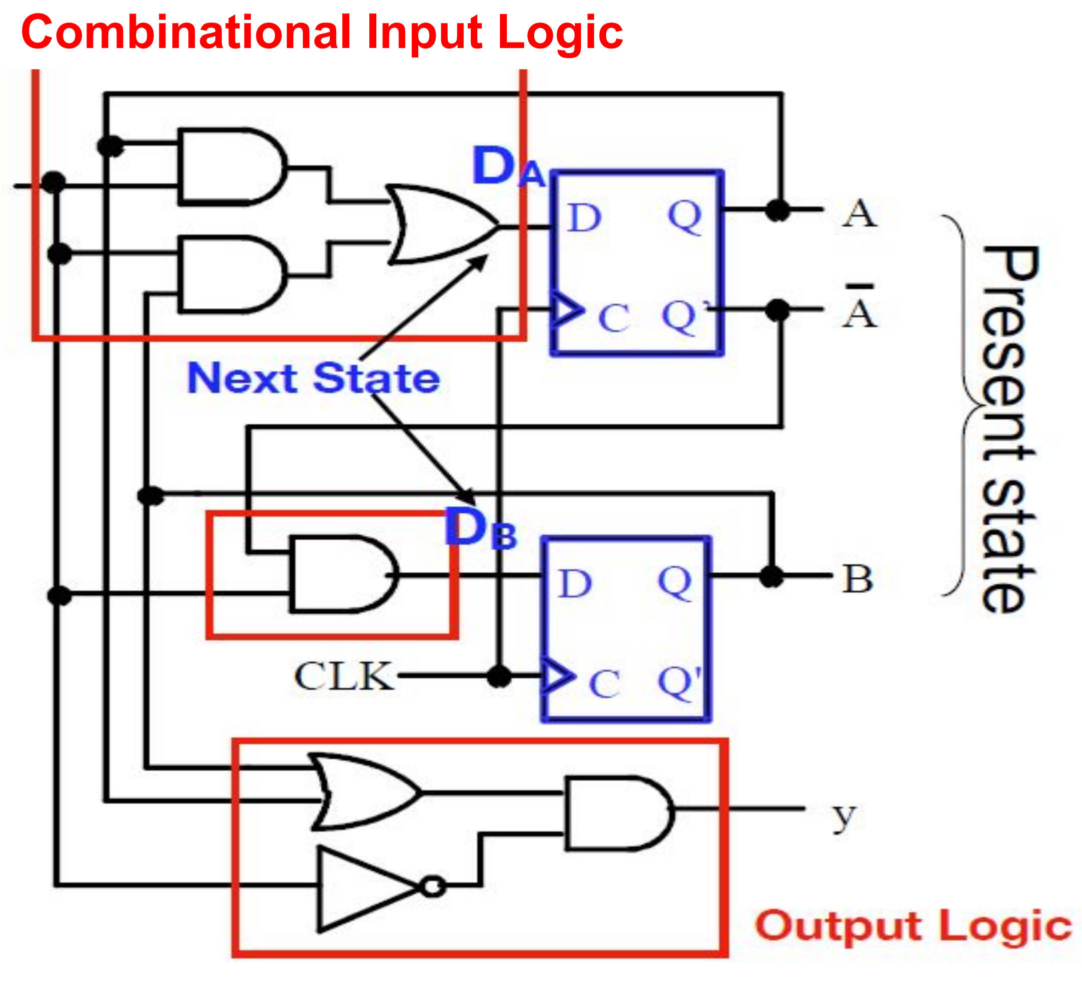
- **Input:** $x(t)$
- **Output:** $y(t)$
- **Present State:** $A(t), B(t)$

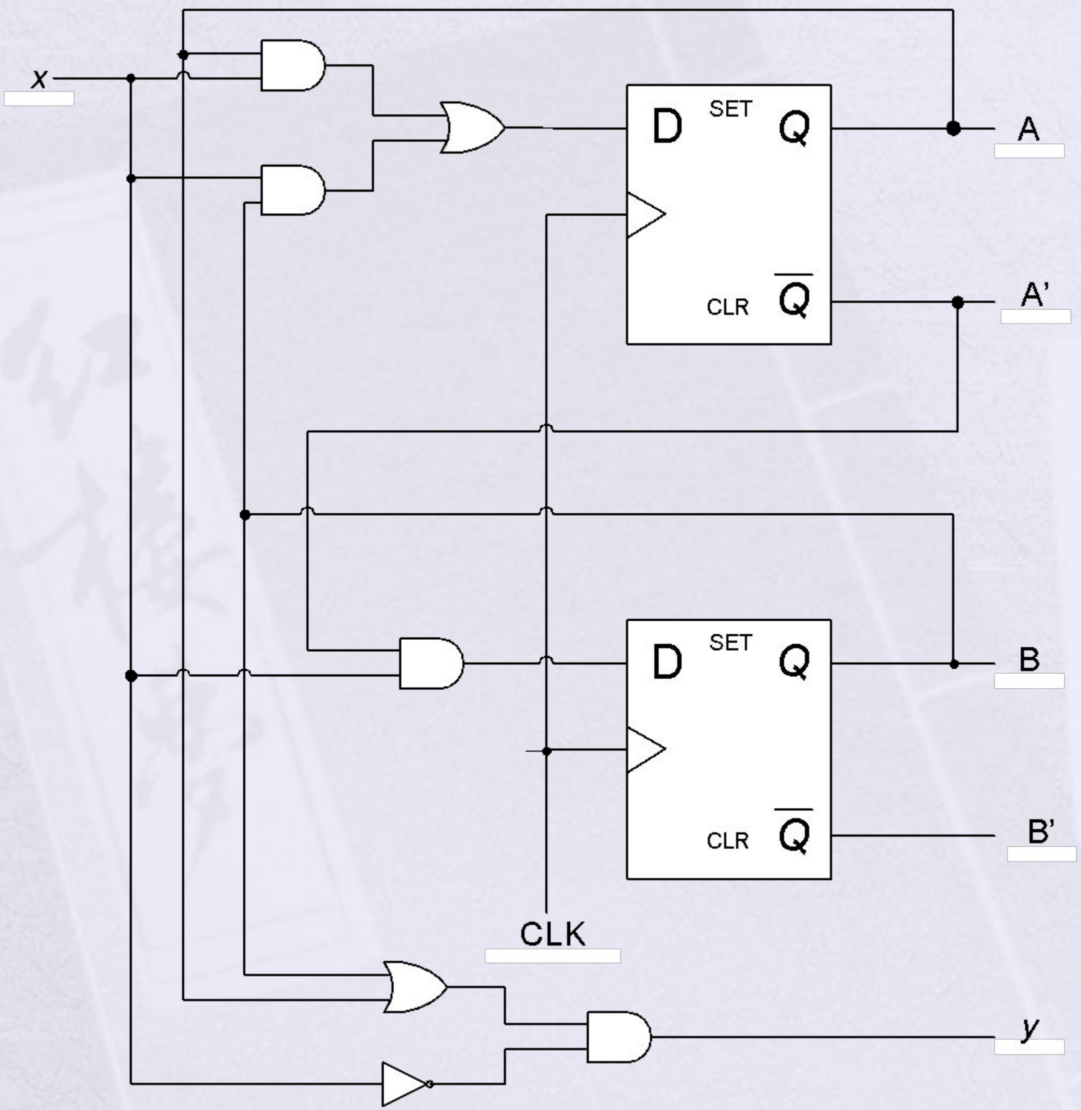
  Example: AB=01, AB=10

- **Next State:**

$$D_A(t), D_B(t)$$

$$= A(t+1), B(t+1)$$





Since D input of the flip flop determines the next state, we can write a set of state equation as

$$\underline{A(t+1) = Ax + Bx}$$

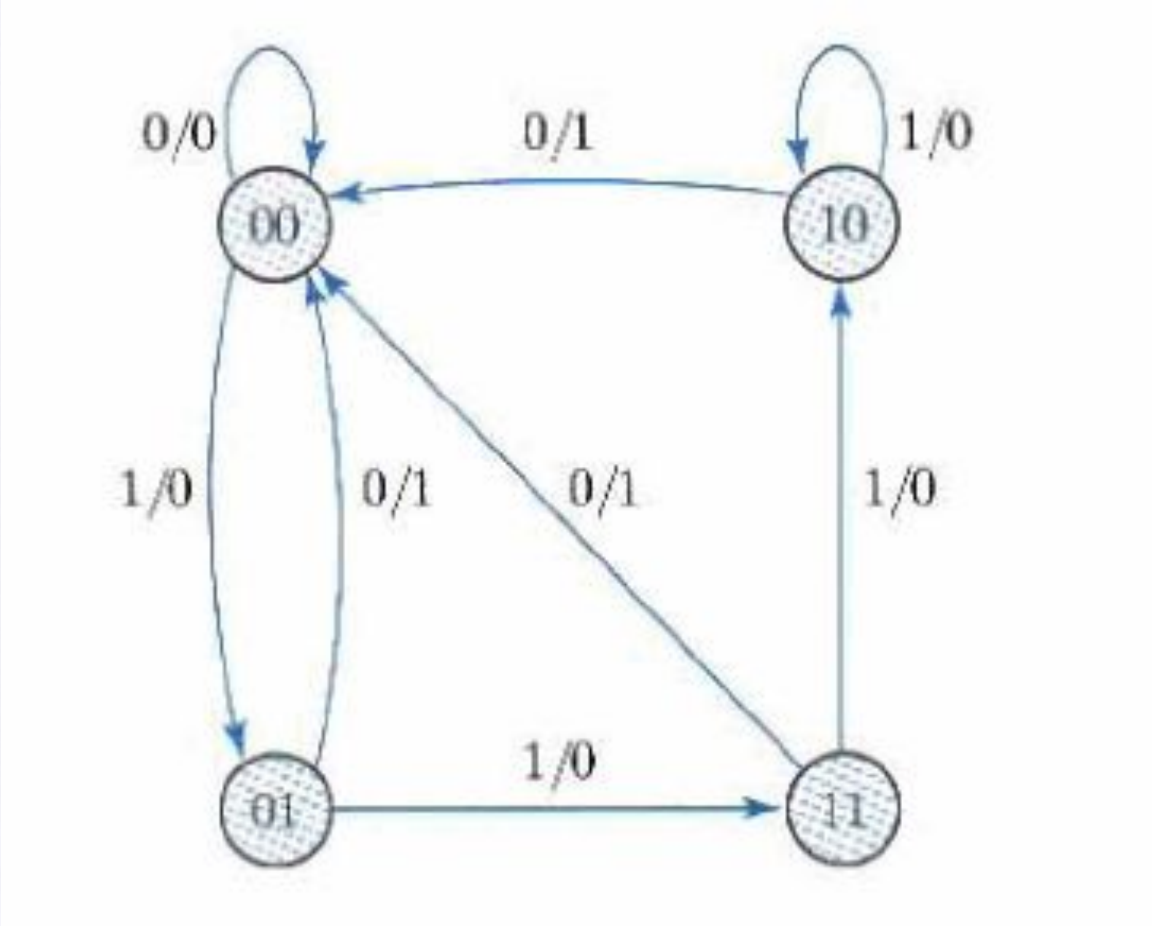
$$\underline{B(t+1) = A'x}$$

And

$$\underline{y(t) = [A(t) + B(t)]\, x'(t)}$$

$$\underline{y = (A + B)x'}$$

Present State		Input	Next State		Output
A	B	x	A	B	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0



Circle = Present State
States= Input/output

Analysis with D Flip Flop

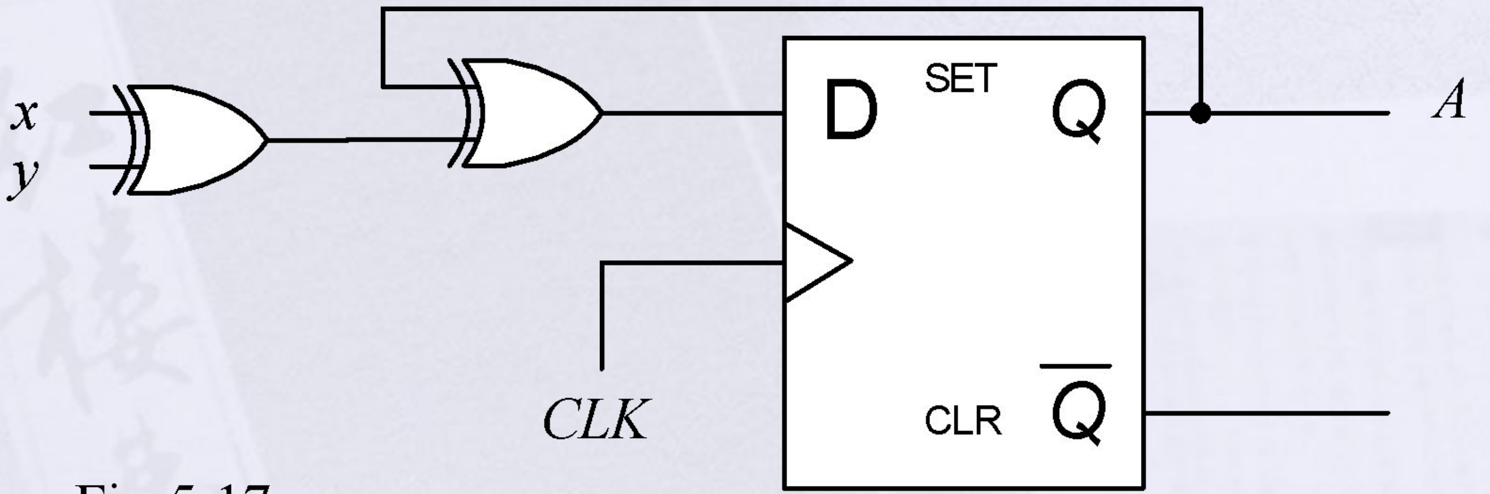


Fig 5-17

Analysis with D Flip Flop

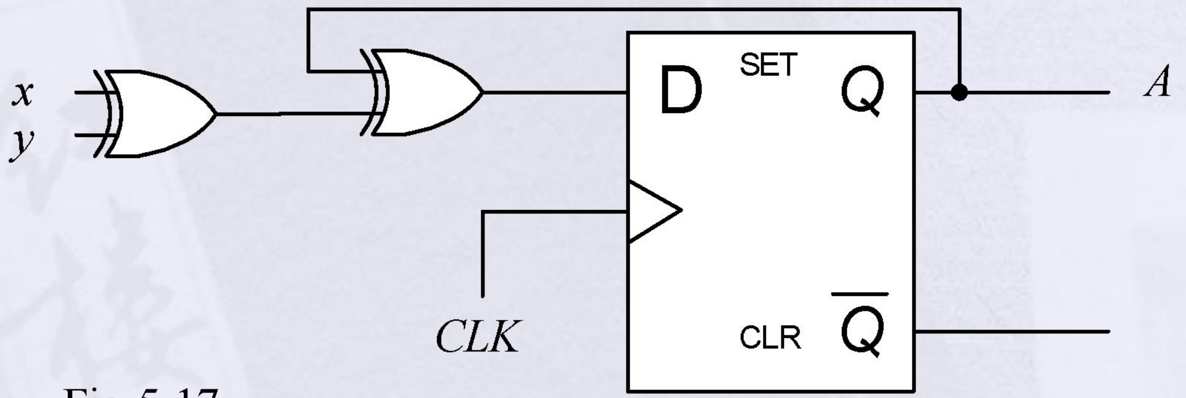


Fig 5-17

- $D_A = A \oplus x \oplus \boldsymbol{y}$
- $A(t+1) = A \oplus x \oplus \boldsymbol{y}$

Analysis with D Flip Flop

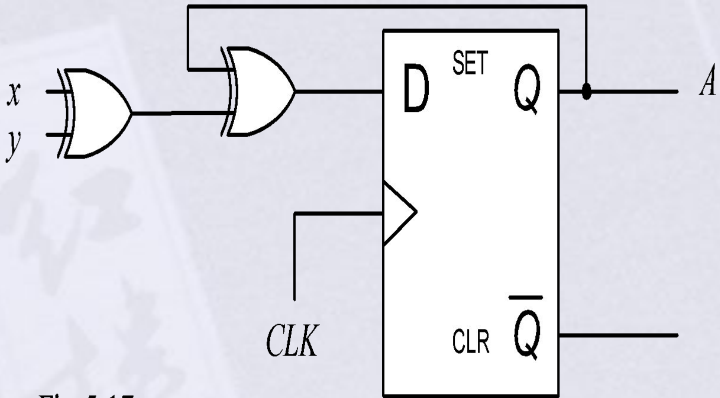
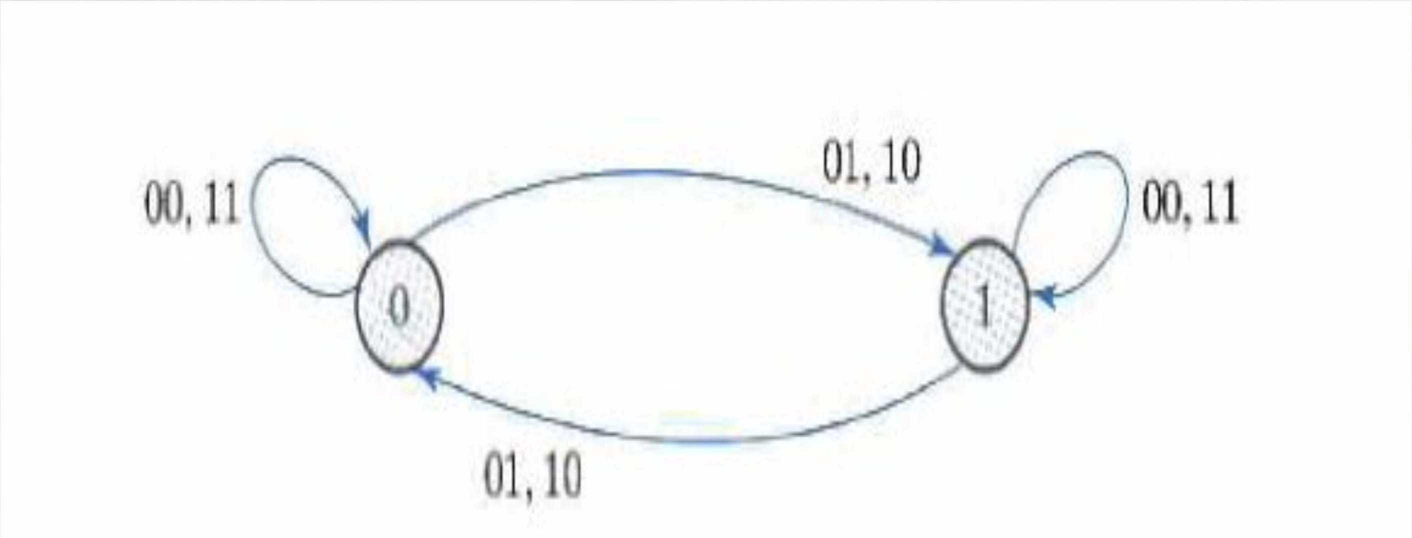


Fig 5-17

Present State	Input		Next State
A	x	y	A
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

- $D_A = A \oplus x \oplus y$
- $A(t+1) = A \oplus x \oplus y$

Analysis with D Flip Flop



Circle = Present State
States = Input

Analysis with D Flip Flop

- A sequential circuit with two D flip-flops A and B, one input x, and one output y is specified by the following next-state and output equations

$$\underline{A(t+1) = Ax + Bx}$$
$$\underline{B(t+1) = A'x}$$
$$\underline{y = (A + B)x'}$$

a) Draw the logic diagram of the circuit.

b) List the state table for the sequential circuit.

c) Draw the corresponding state diagram.

Analysis with JK Flip Flop

- The next-state value of a sequential circuit that uses J-K flip-flops can be derived using the following procedure.
  1. determine the flip-flop input equation in terms of present state and input variable.
  2. List the binary values of each input equation
  3. Use the corresponding flip-flop characteristic table to determine the next values in the state table.

Analysis with JK Flip Flop

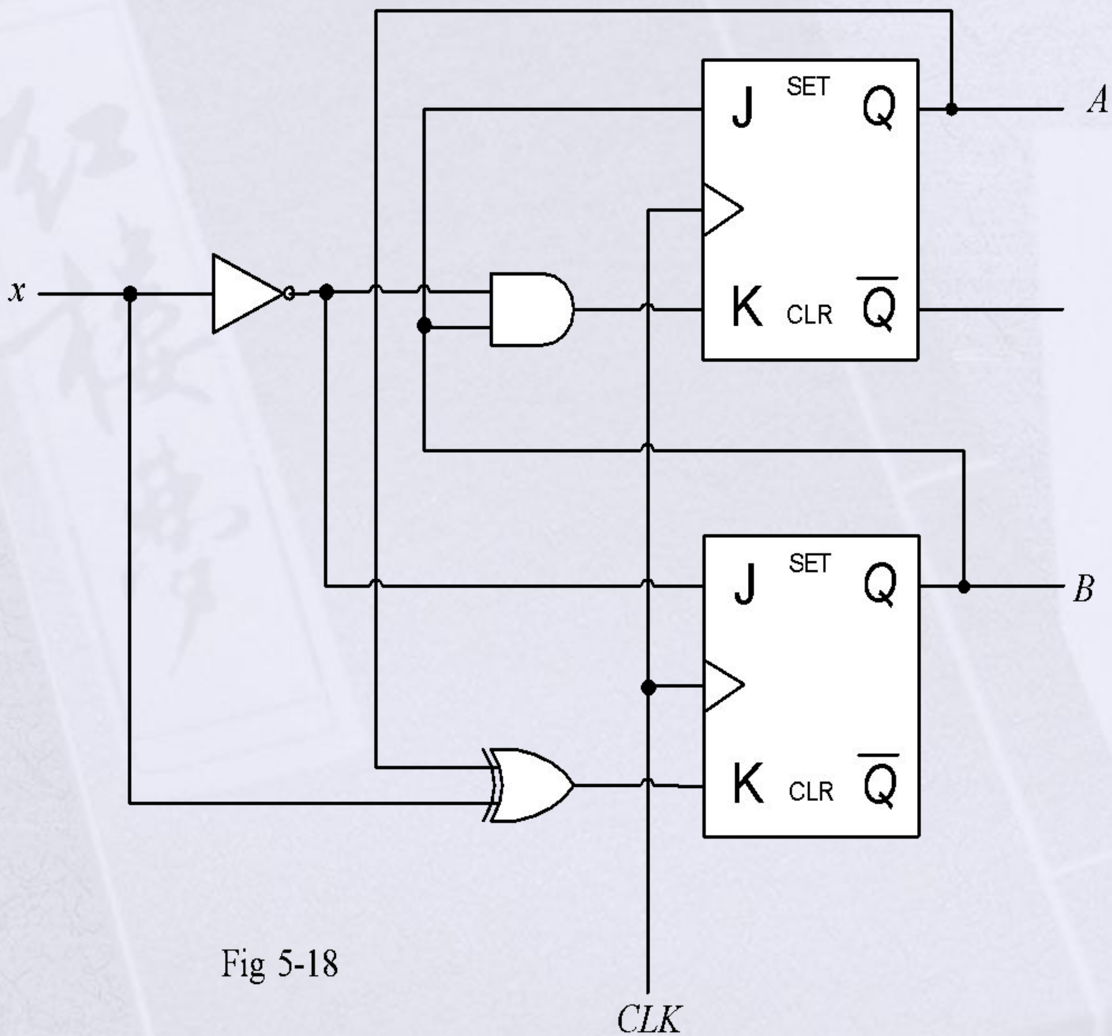


Fig 5-18

$J_A = B$

$K_A = Bx'$

$J_B = x'$

$K_B = A'x + Ax'$
$= A \oplus x$

Analysis with JK Flip Flop

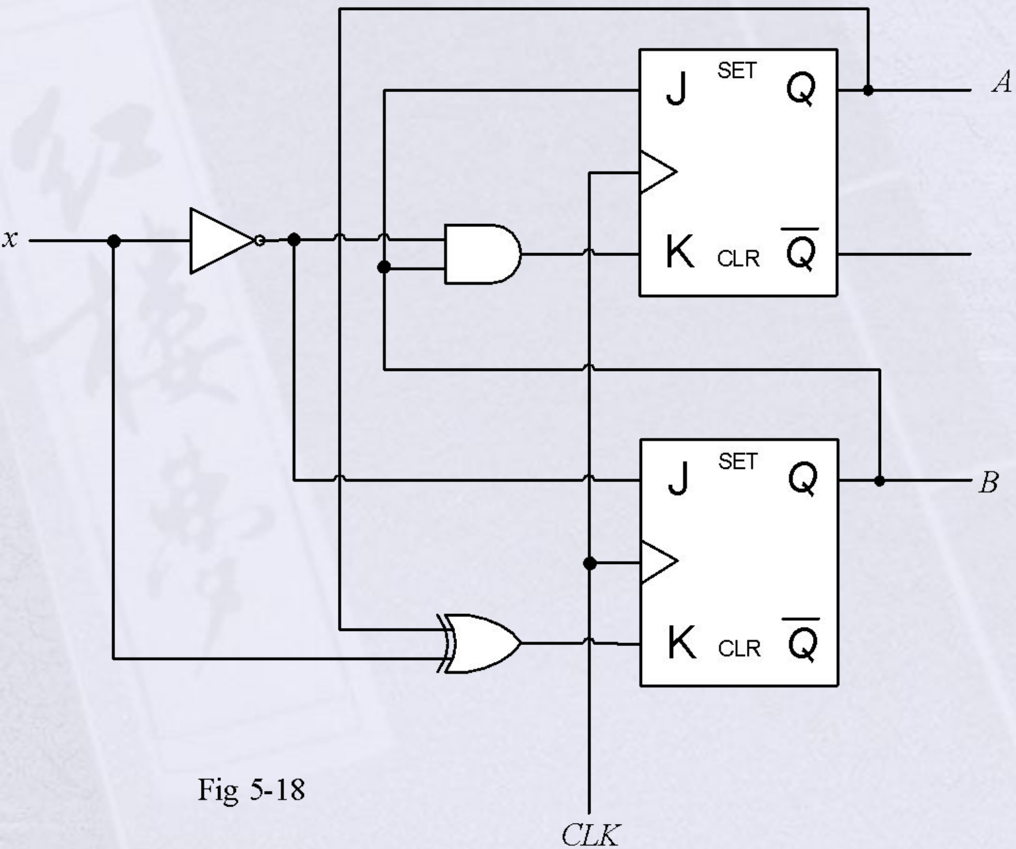


Fig 5-18

$J_A = B$ $\quad\quad$ $K_A = Bx'$

$J_B = x'$ $\quad\quad$ $K_B = A'x + Ax'$

$= A \oplus x$

$A(t+1) = JA' + K'A$

$= A'B + (Bx')'A$

$= A'B + AB' + Ax$

$B(t+1) = JB' + K'B$

$= B'x' + (A \oplus x)'B$

$= B'x' + ABx + A'Bx'$

Analysis with JK Flip Flop

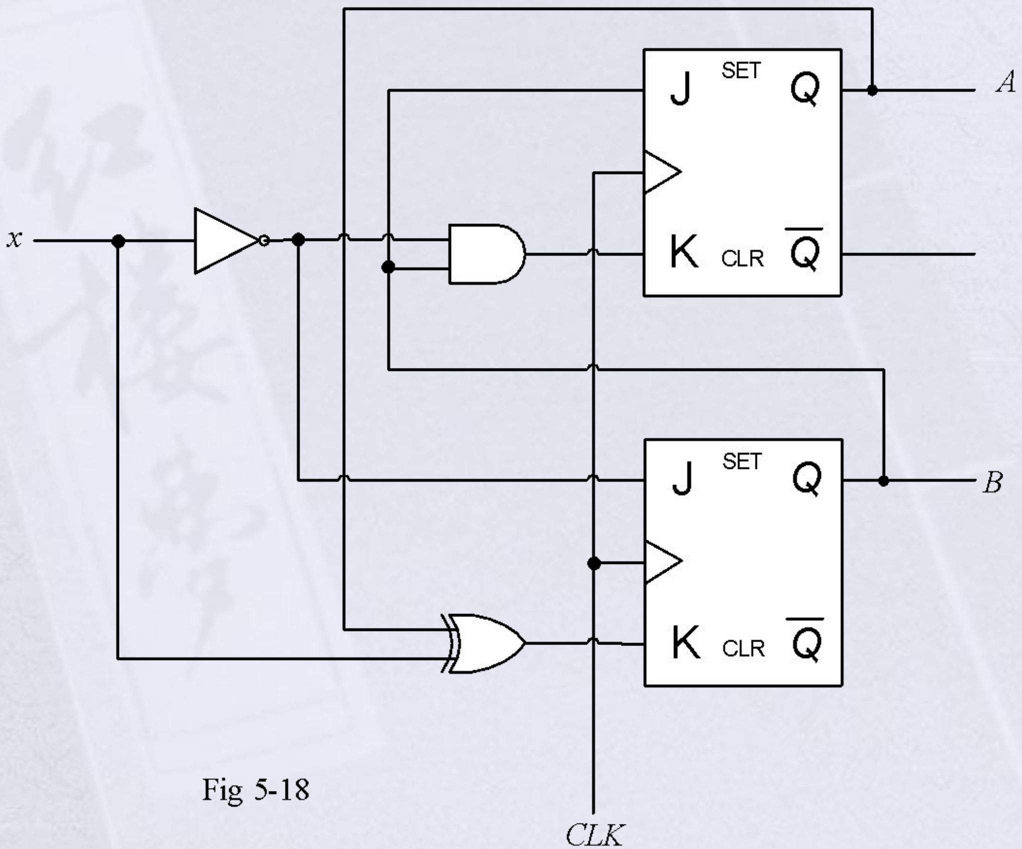


Fig 5-18

$A(t+1) = A'B + AB' + Ax$

$B(t+1) = B'x' + ABx + A'Bx'$

Present State		Input	Next States	
A	B	x	A	B
0	0	0	0	1
0	0	1	0	0
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	0	0
1	1	1	1	1

Analysis with JK Flip Flop

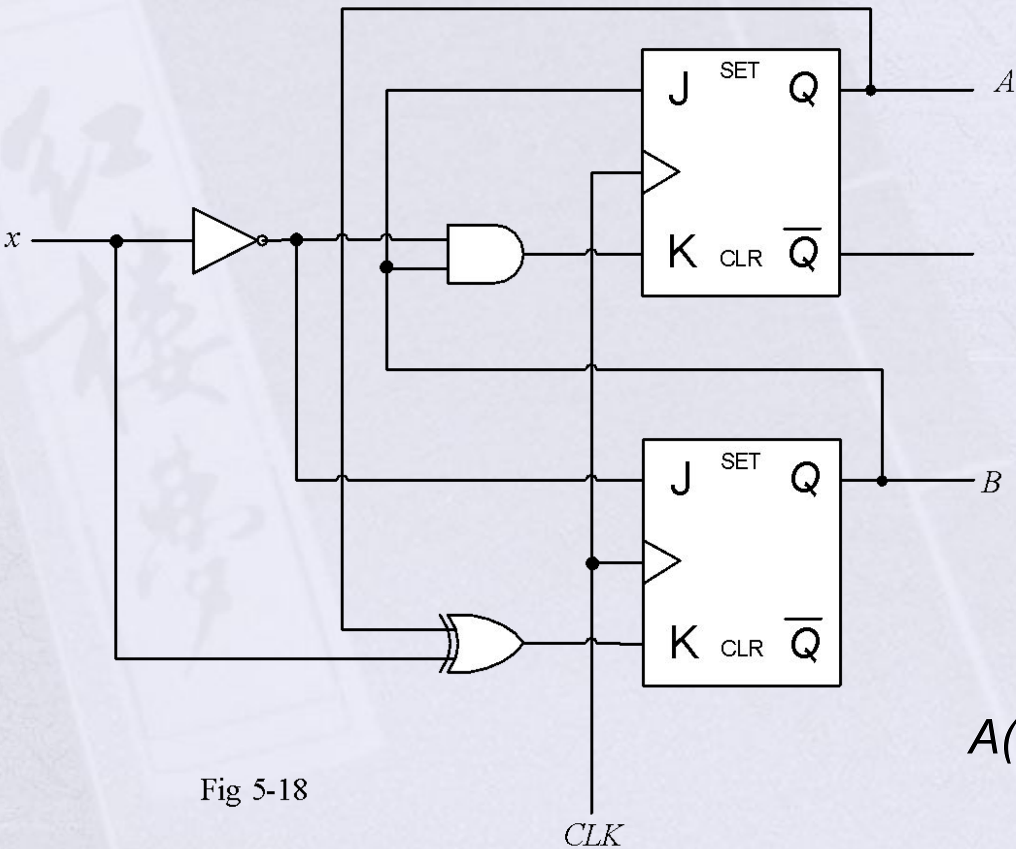


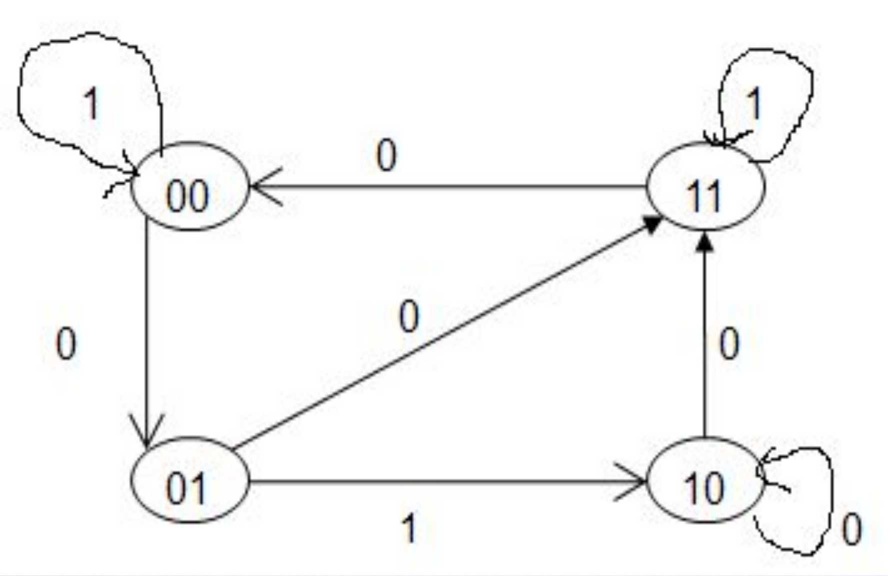
Fig 5-18

Present State		Input		
A	B	x	A	B
0	0	0	0	1
0	0	1	0	0
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	0	0
1	1	1	1	1

$$A(t+1) = JA' + K'A$$
$$= A'B + AB' + Ax$$

$$B(t+1) = JB' + K'B$$
$$= B'x' + ABx + A'Bx'$$

Analysis with JK Flip Flop



Circle = Present State
States= Input/output

Present State		Input		
A	B	x	A	B
0	0	0	0	1
0	0	1	0	0
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	0	0
1	1	1	1	1

Analysis with JK Flip Flop

A sequential circuit has two JK flipflop A and B and one input x. The circuit is described by the following flip-flop input equations

$$J_A = B \qquad K_A = B'x$$
$$J_B = x' \qquad K_B = A'x + Ax'$$
$$= A \oplus x$$

(a) Draw the logic diagram of the circuit.
(b) Derive the state equations for A and B.
(c) Tabulate the state table.
(d) Draw the state diagram of the circuit.

Analysis with T Flip-Flops:

- Same procedure as J-K flip-flops.

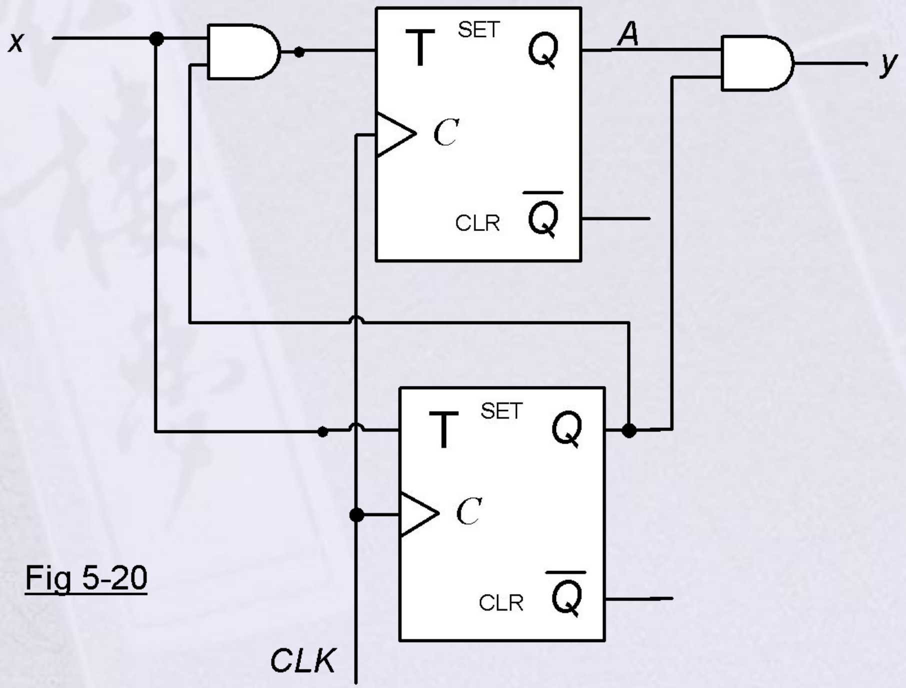


Fig 5-20

Analysis with T Flip-Flops:

- Same procedure as J-K flip-flops.

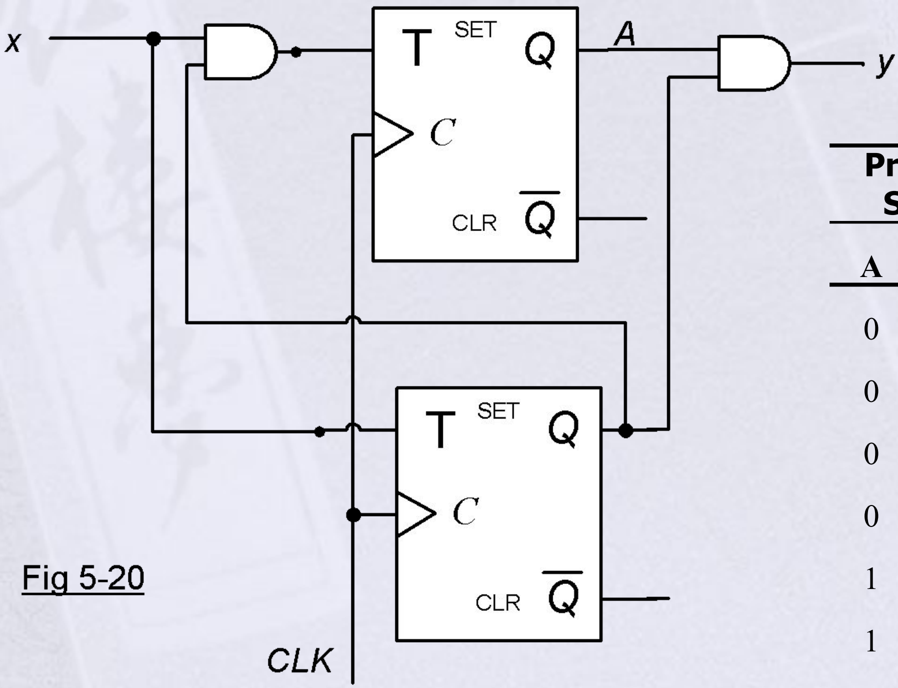


Fig 5-20

Present State		Input	Next State		Output
A	B	x	A	B	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	1

Analysis with T Flip-Flops:

- Same procedure as J-K flip-flops.

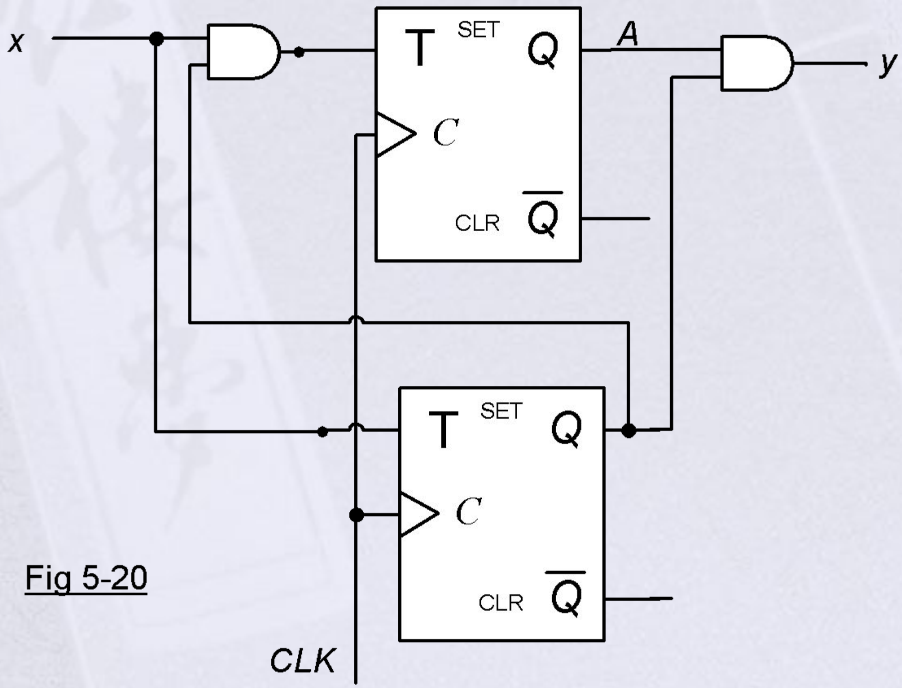


Fig 5-20

Analysis with T Flip-Flops:

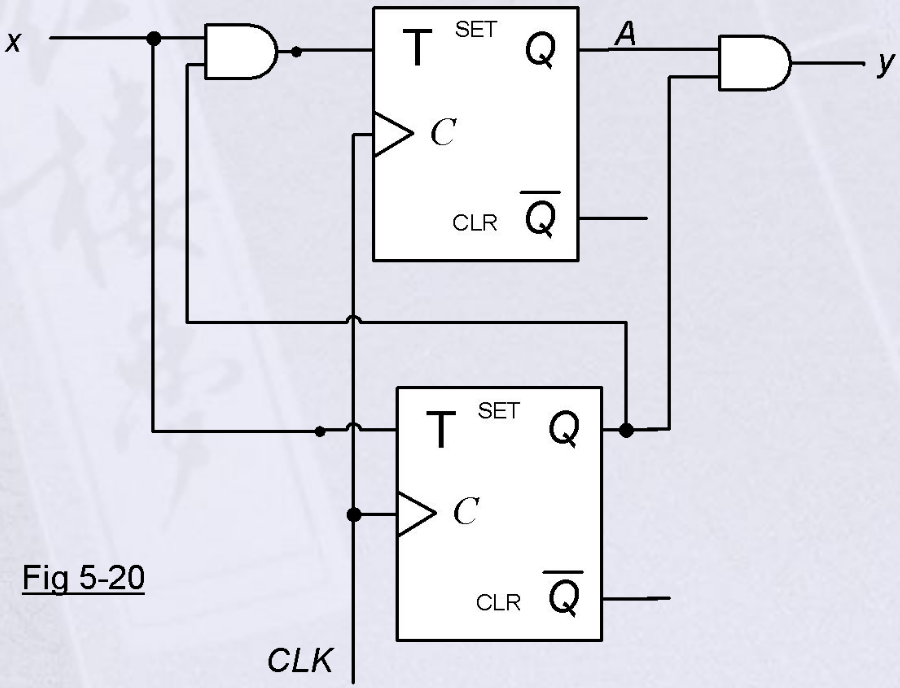


Fig 5-20

- $Q\,(t+1) = T\;Q = T'Q + TQ'$
- $T_A = Bx,\ T_B = X,\ y{=}AB$
- $A(t+1) = (Bx)'A + (Bx)A'$
  $= AB' + Ax' + A'Bx$
- $B(t+1) = x \oplus B$

Analysis with T Flip-Flops:

- Same procedure as J-K flip-flops.

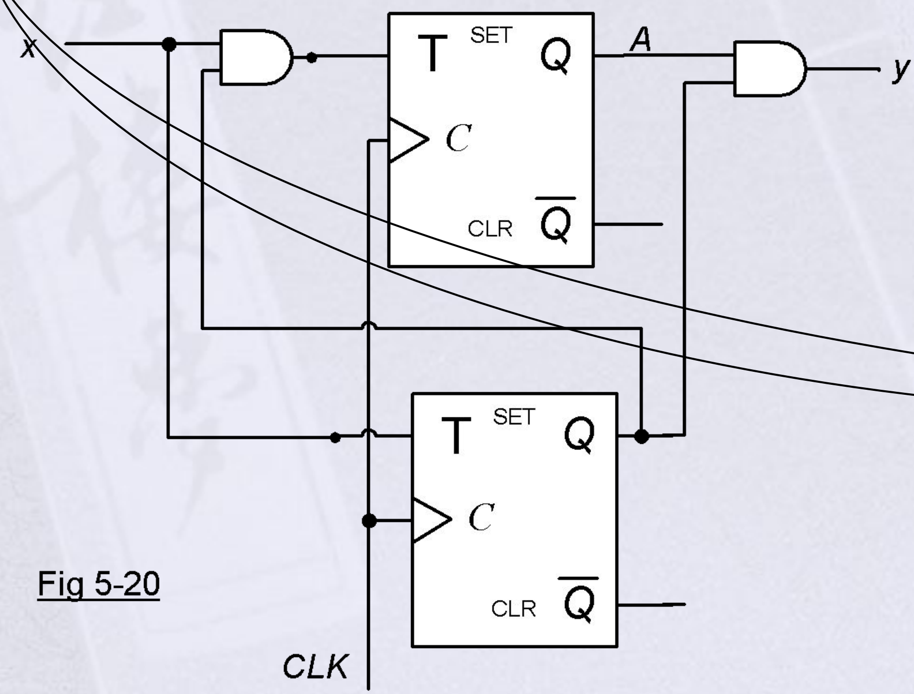


Fig 5-20

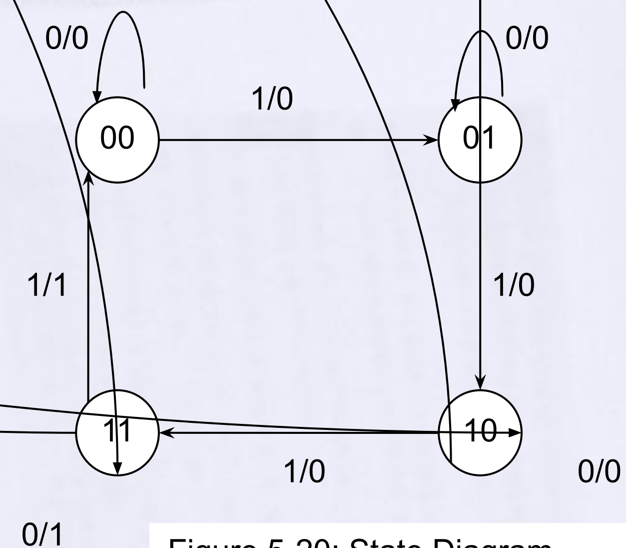


Figure 5-20: State Diagram

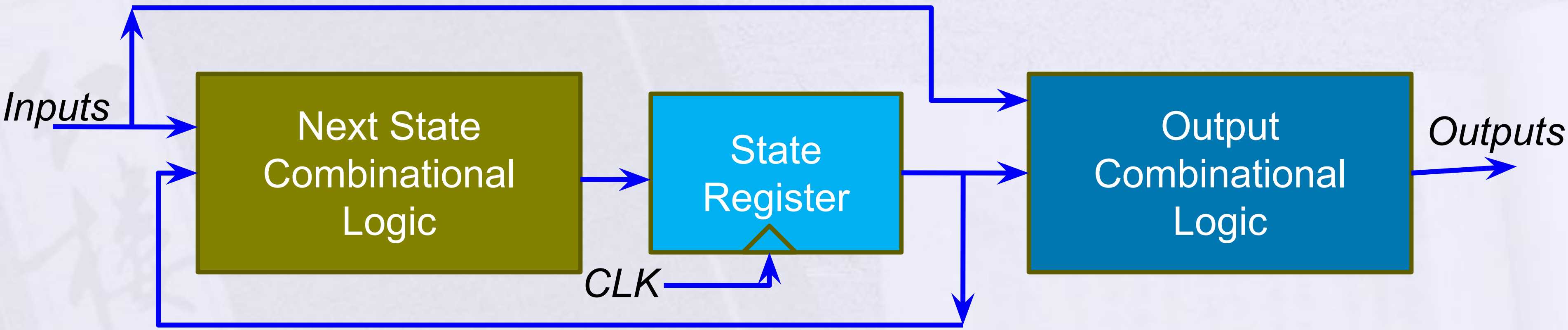
Circle = Present State
States= Input/output

MEALY AND MOORE MODEL

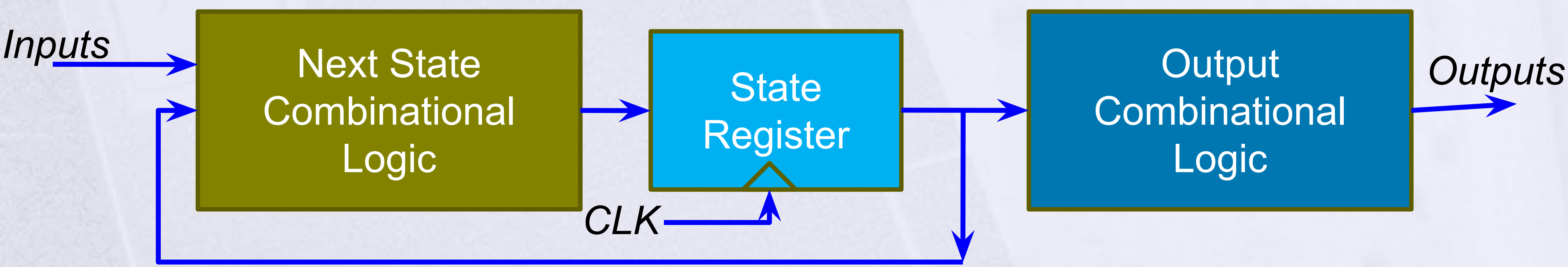
Mealy and Moore Model

- Meelay Model:
  - Output is the function of both the inputs and the present states
  - Outputs = f(inputs, state)
- Moore Model
  - Output is the function of present states only.
  - Output combinational logic is not dependent on the input.
  - Outputs = f(state)

Mealy and Moore Model

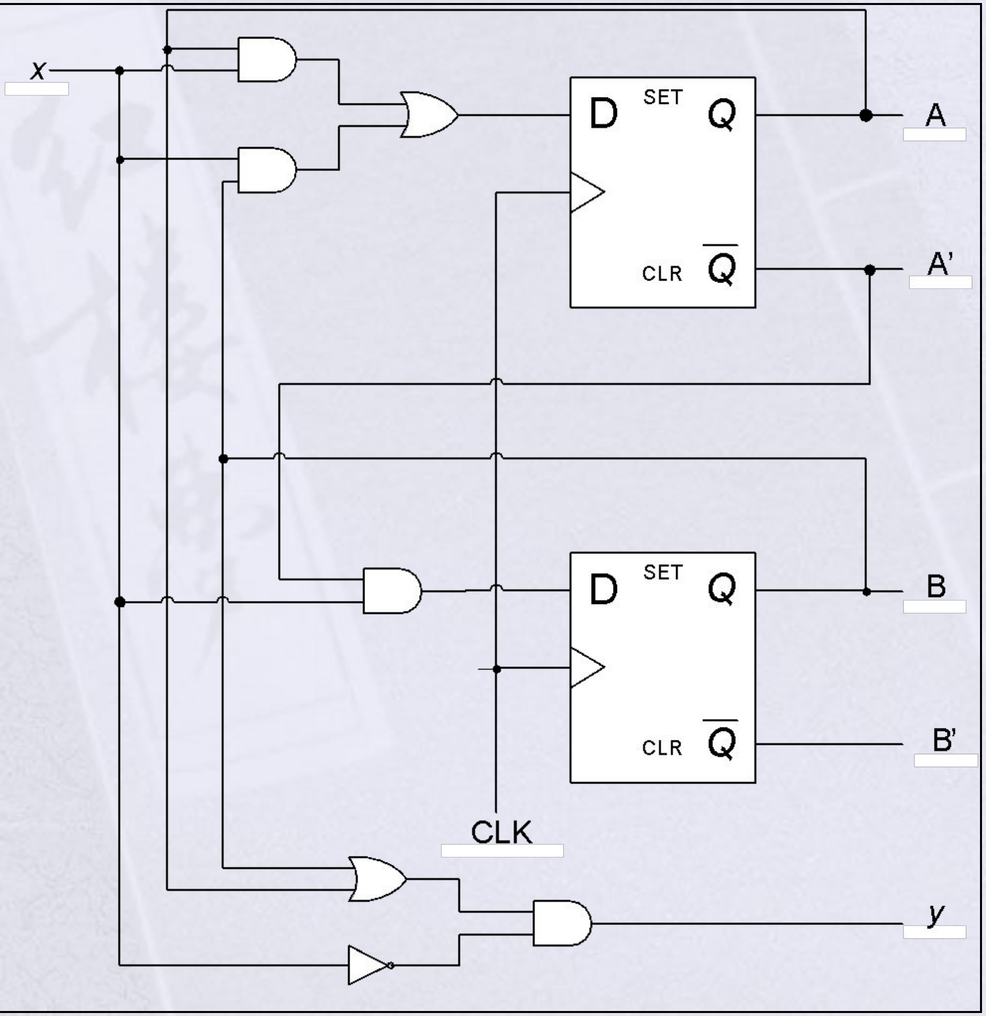


Meelay Machine



Moore Machine

Mealy and Moore Model



Mealy Model

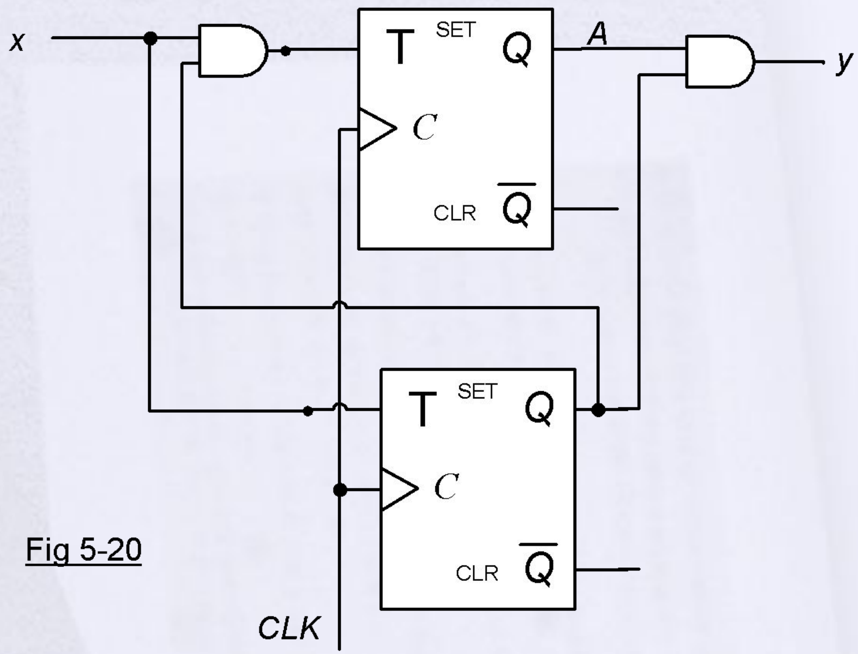
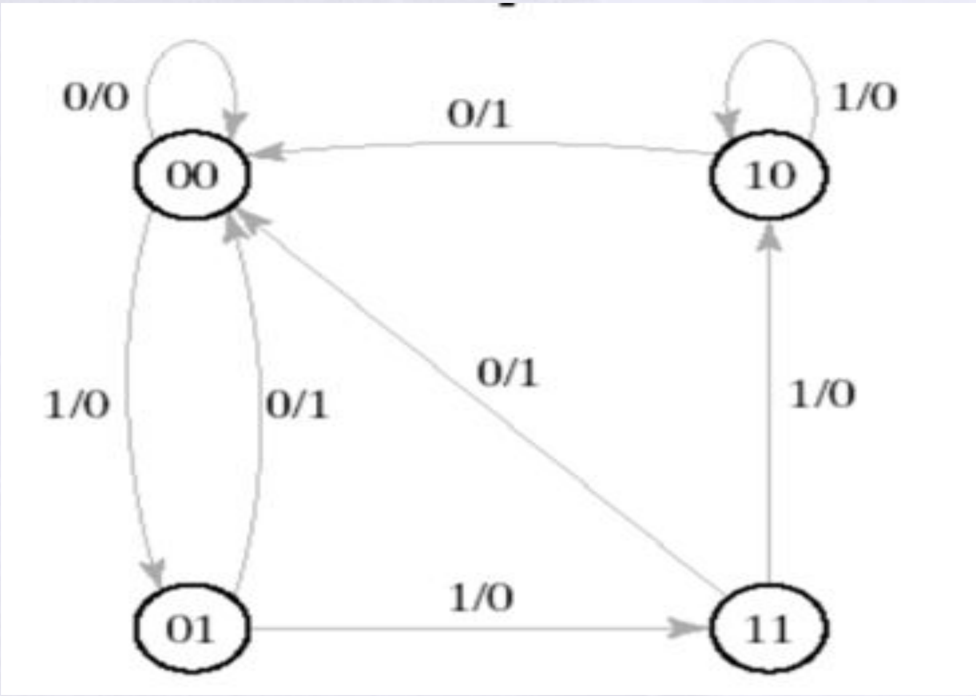


Fig 5-20

Moore Model

Meelay Model

- Output is the function of both the inputs and the present states



Moore Model

- Outputs are put inside the circle

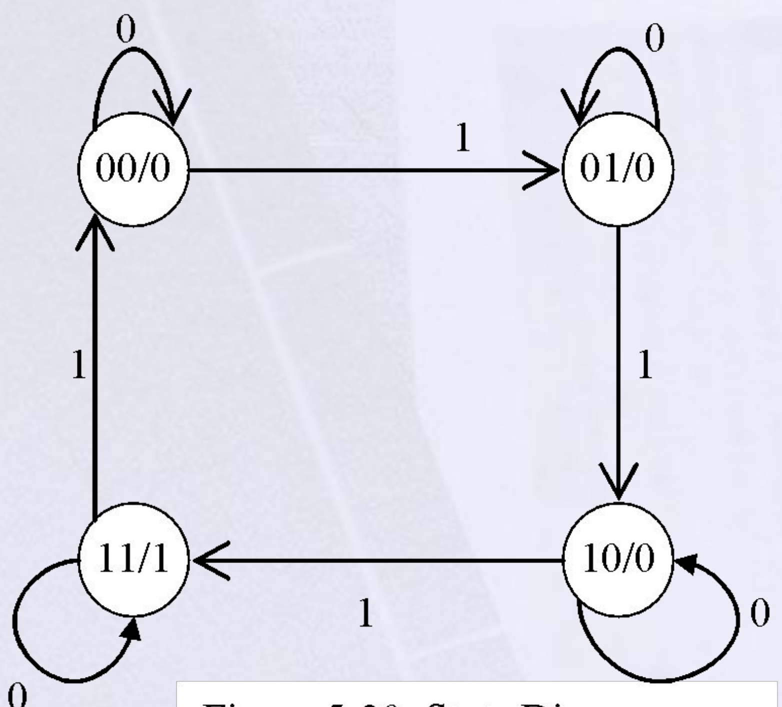


Figure 5-20: State Diagram

STATE REDUCTION

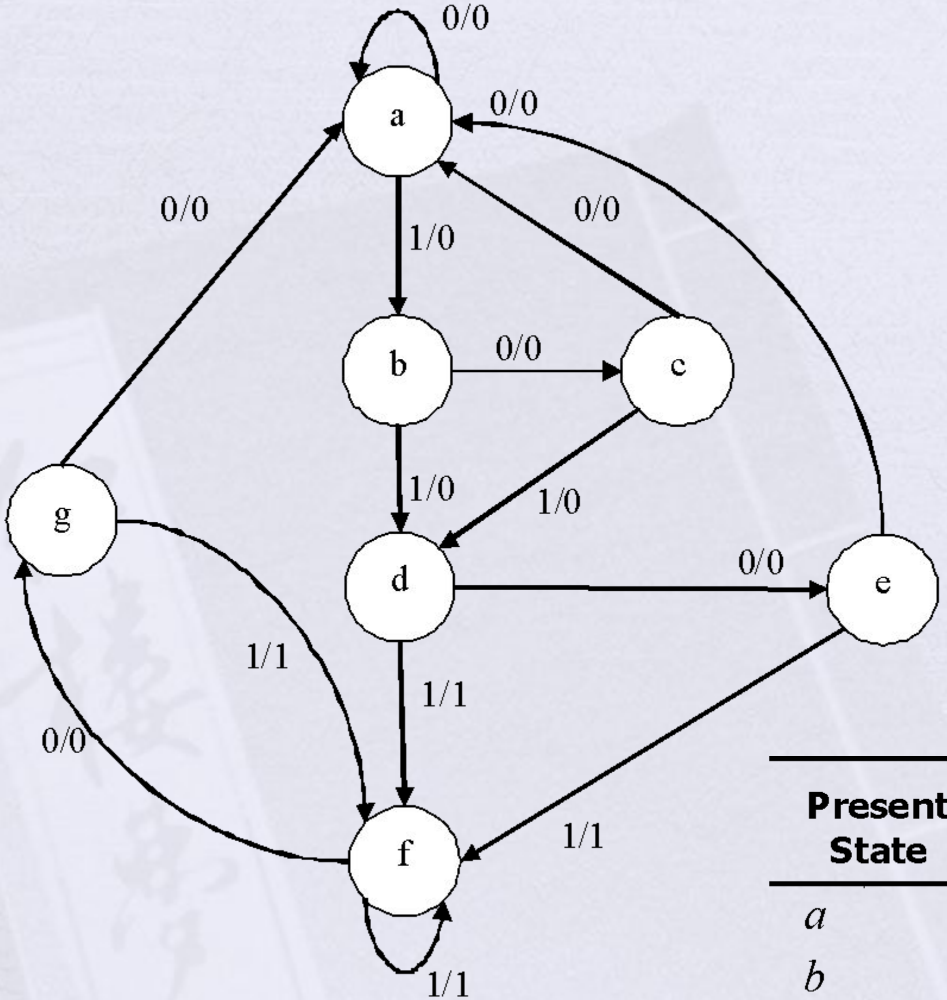
State Reduction

- Reduce the number of states(flip flops) of the circuit without altering the input and output relationship.
- From the state table, draw the state diagram (modified)

State Reduction

When are the two states equivalent?

- Two states are said to be equivalent if, for each of the set of inputs, they give exactly the same output and send the circuit either to the same state or to an equivalent state.



Present State	Next State		Output	
	x = 0	x = 1	x = 0	x = 1
a	*a*	*b*	0	0
b	*c*	*d*	0	0
c	*a*	*d*	0	0
d	*e*	*f*	0	1
e	*a*	*f*	0	1
f	*g*	*f*	0	1
g	*a*	*f*	0	1

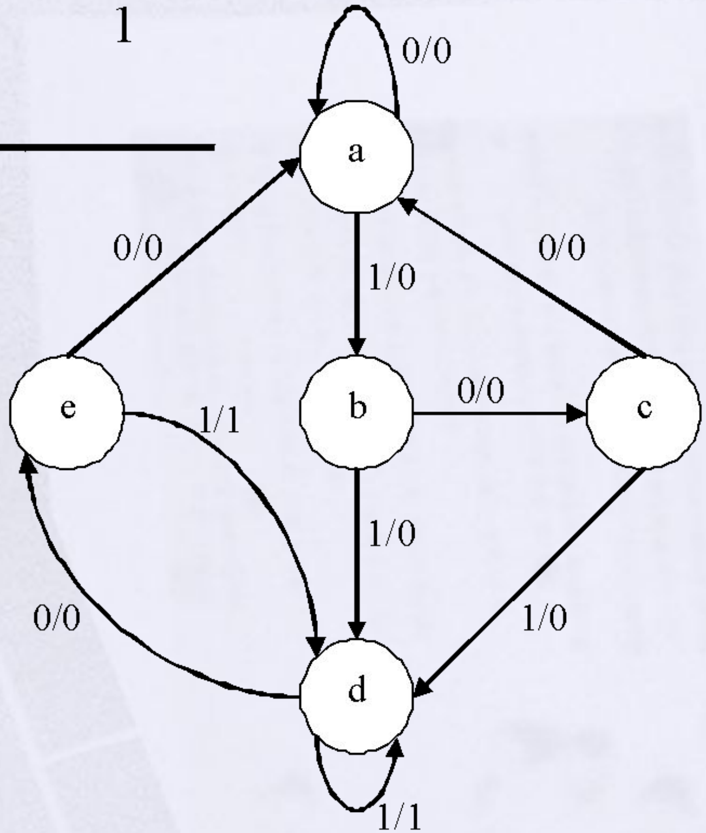
Present State	Next State		Output	
	x = 0	x = 1	x = 0	x = 1
a	*a*	*b*	0	0
b	*c*	*d*	0	0
c	*a*	*d*	0	0
d	*e*	*f*	0	1
e	*a*	*f*	0	1
f	*g*	*f*	0	1
~~*g*~~	~~*a*~~	~~*f*~~	~~0~~	~~1~~

Present State	Next State		Output	
	x = 0	x = 1	x = 0	x = 1
a	*a*	*b*	0	0
b	*c*	*d*	0	0
c	*a*	*d*	0	0
d	*e*	*f*	0	1
e	*a*	*f*	0	1
f	*e*	*f*	0	1

Present State	Next State		Output	
	x = 0	x = 1	x = 0	x = 1
a	*a*	*b*	0	0
b	*c*	*d*	0	0
c	*a*	*d*	0	0
d	*e*	*f*	0	1
e	*a*	*f*	0	1
f	*e*	*f*	0	1

Present State	Next State		Output	
	x = 0	x = 1	x = 0	x = 1
a	*a*	*b*	0	0
b	*c*	*d*	0	0
c	*a*	*d*	0	0
d	*e*	*d*	0	1
e	*a*	*d*	0	1

Present State	Next State		Output	
	x = 0	x = 1	x = 0	x = 1
a	*a*	*b*	0	0
b	*c*	*d*	0	0
c	*a*	*d*	0	0
d	*e*	*d*	0	1
e	*a*	*d*	0	1

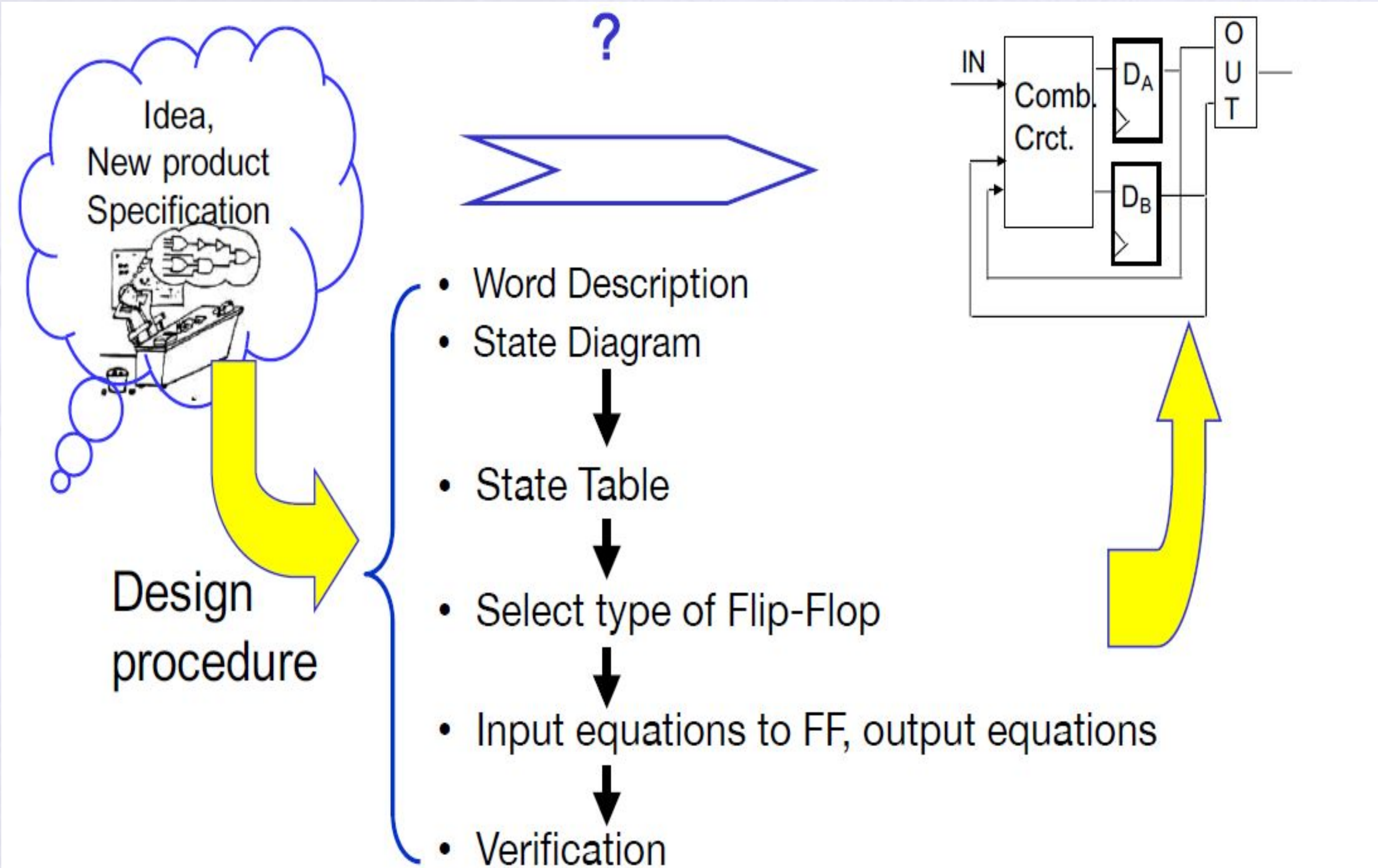


DESIGN WITH FLIP FLOPS

Design Procedure

- From the word description of specifications of desired operation, derive a state diagram for the circuit.
- Assign binary value to the states.
- Obtain the binary coded state table
- Choose the type of flip flops to be used
- Derive the simplified flip-flop-input equations and output equations
- Draw the logic diagram

Design Procedure



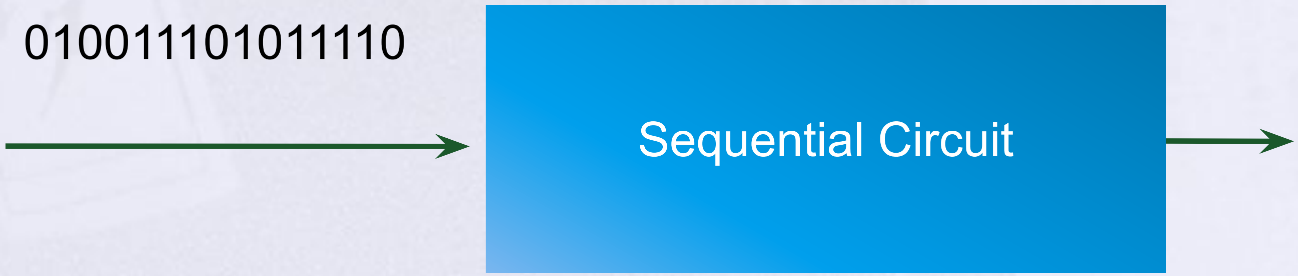
Example

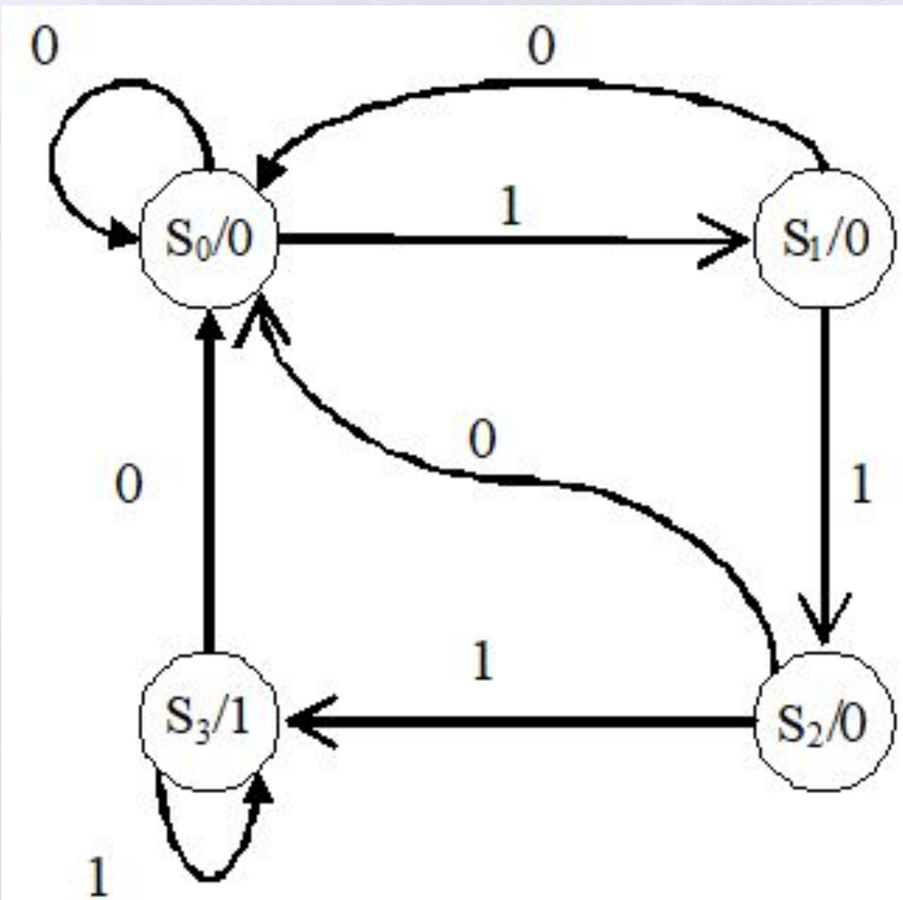
- **Description/Specification**
- Design a circuit that detects three or more consecutive 1's in a string of bits coming through an input line

Example

- **Description/ Specification**

- Design a circuit that detects three or more consecutive 1's in a string of bits coming through an input line





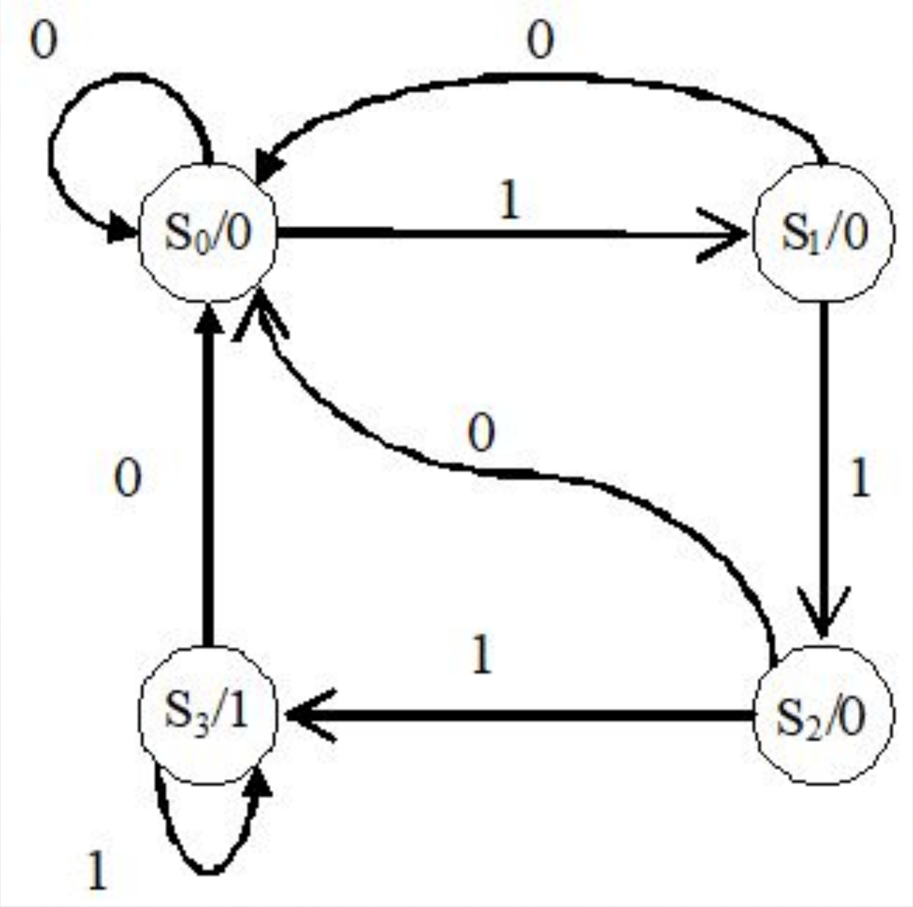
State S_0: zero 1s detected

State S_1: one 1 detected

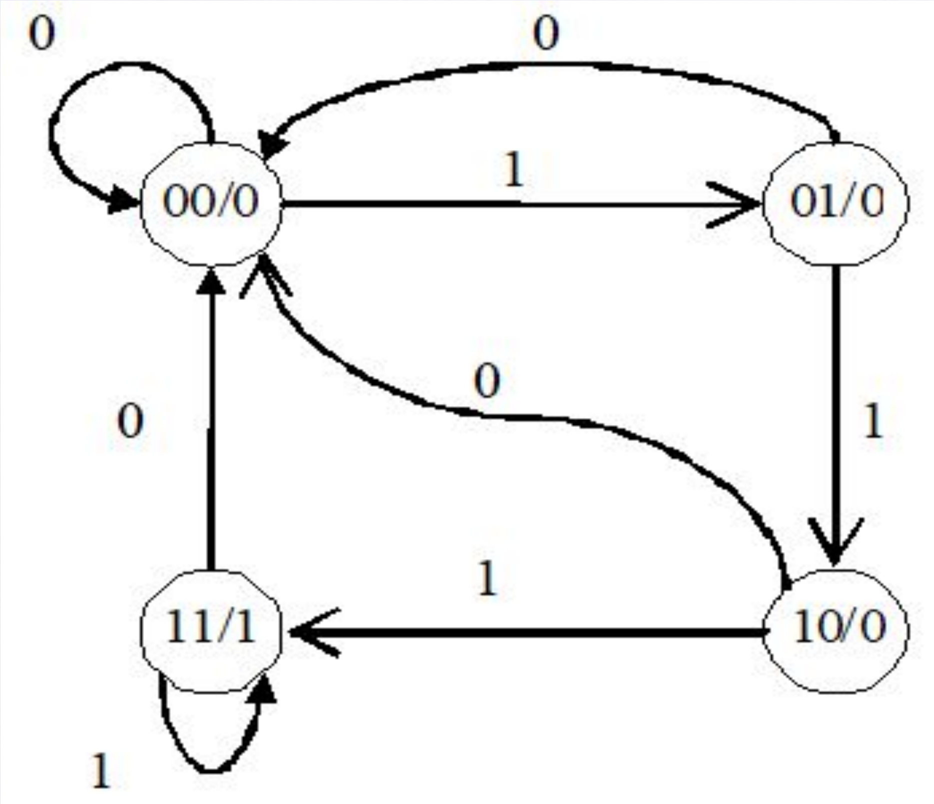
State S_2: two 1s detected

State S_3: three 1s detected

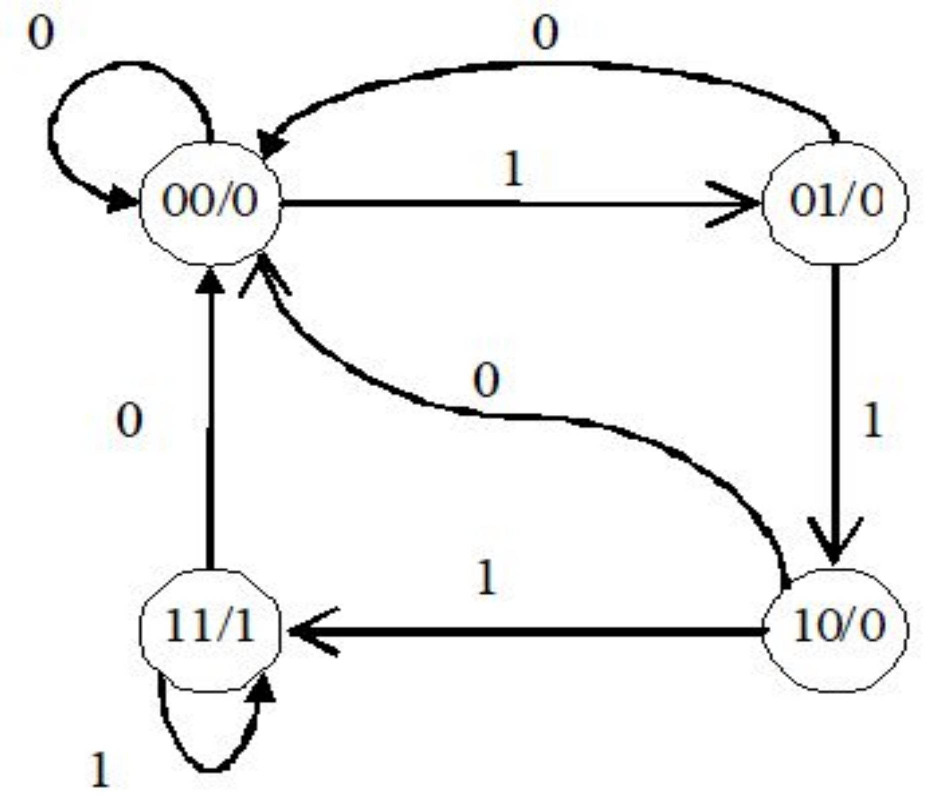
In circle: State of Flip-Flops / Output
Outside: input Value



- Name the states after drawing the state diagram.
- Since there are four states in this diagram, we need two flip flops.
- Hence the states will be 00, 01, 10 and 11; as shown below.



In circle: State of Flip-Flops / Output
Outside: input Value



Present State		Input	Next State		Output
A	B	x	A	B	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	1	1	0
1	1	0	0	0	1
1	1	1	1	1	1

- Present state indicates current value of flip flops
- Next state indicates state after next rising clock edge
- Output is current output value

In circle: State of Flip-Flops / Output
Outside: input Value

State Equations

$A(t+1) = D_A(A,B,x) = \sum (3,5,7)$

$B(t+1) = D_B(A,B,x) = \sum (1,5,7)$

$y(A,B,x) = \sum (6,7)$

Present State		Input	Next State		Output
A	B	x	A	B	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	1	1	0
1	1	0	0	0	1
1	1	1	1	1	1

State Equations

$A(t+1) = D_A(A,B,x) = \sum(3,5,7)$

$B(t+1) = D_B(A,B,x) = \sum(1,5,7)$

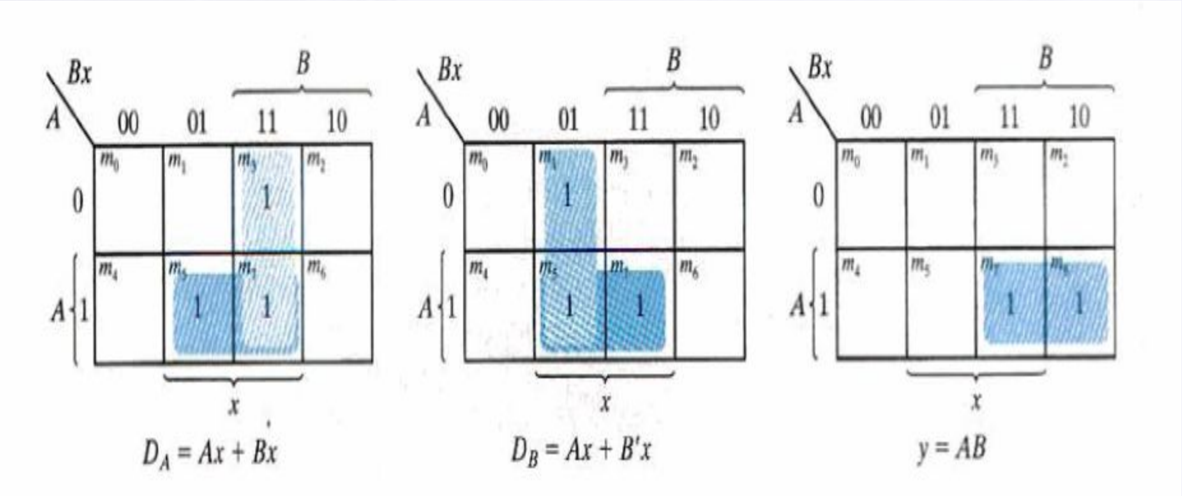
$y(A,B,x) = \sum(6,7)$

By Simplification:

$D_A = Ax + Bx$

$D_B = Ax + B'x$

$y = AB$



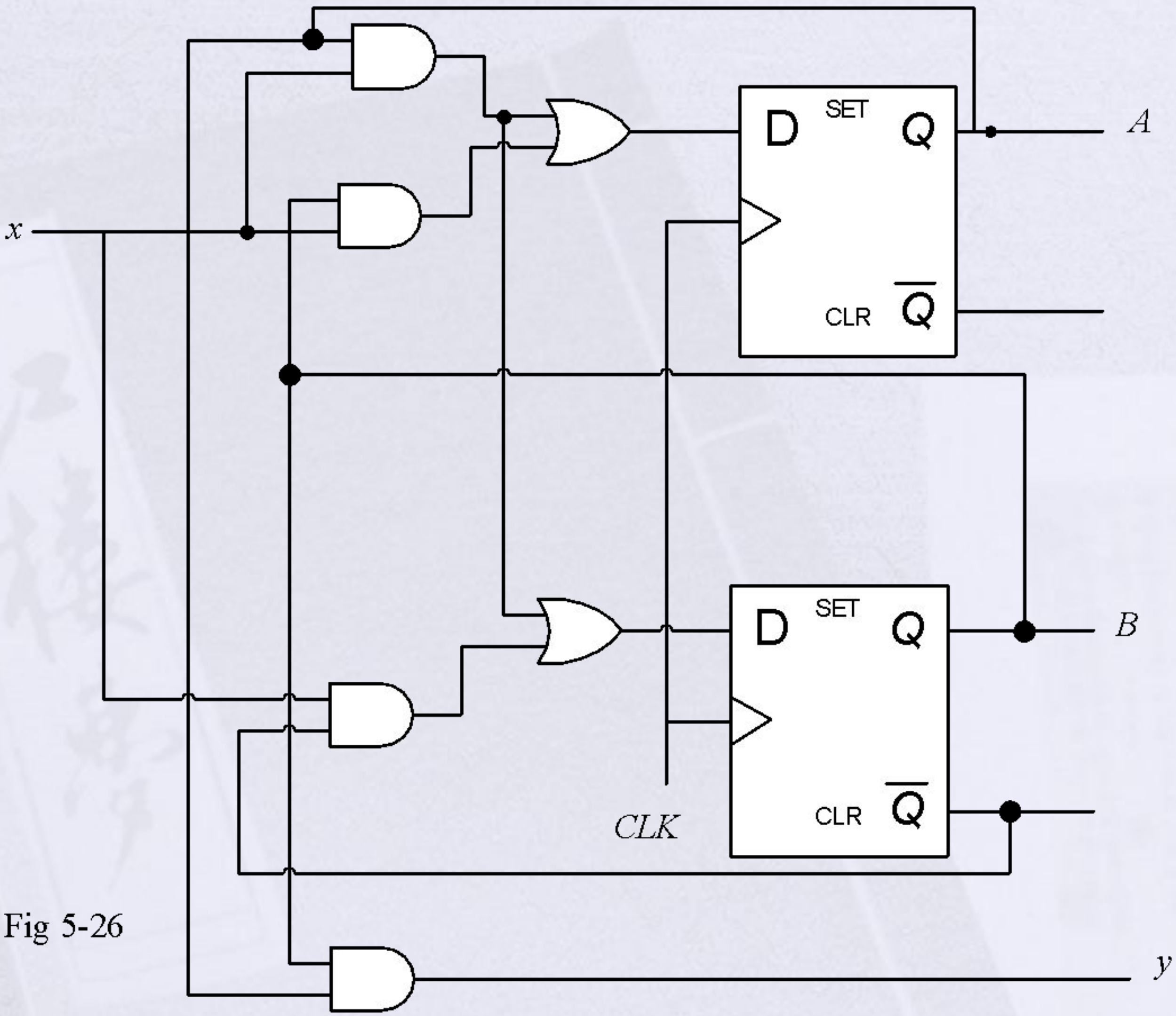


Fig 5-26

$D_A = Ax + Bx$

$D_B = Ax + B'x$

$y = AB$

Question

- Design a sequential circuit with two JK flip-flops A and B and one input x-in.

When x- in = 0, the state of the circuit remains the same except for 01 where it becomes 10. When x-in = 1, the circuit goes through the state transitions from 00 to 01 ,01 to 01,11 to 00, 10 back to 11.

Design with JK-Flip Flop

- Excitation Table

Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

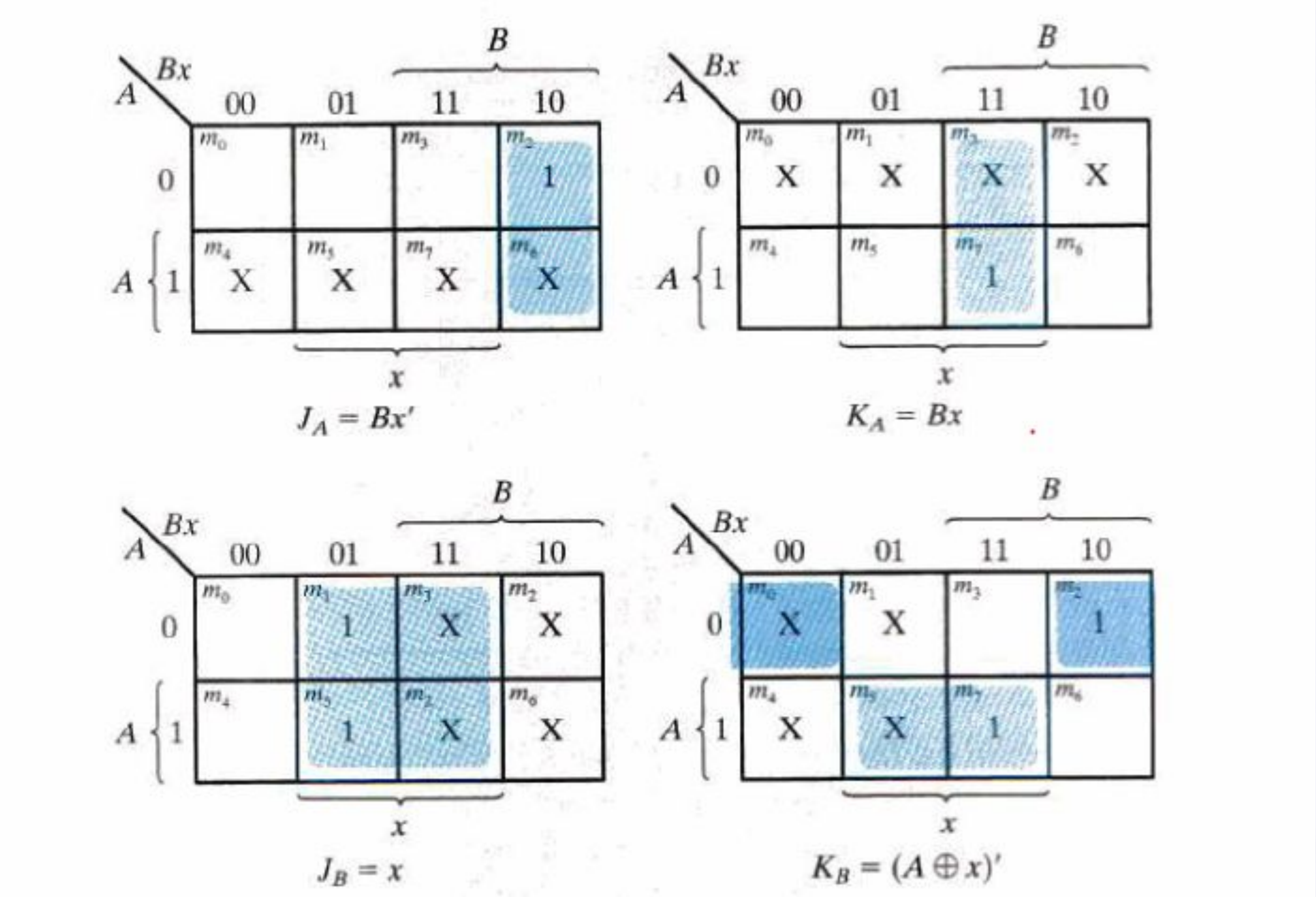
Q(t)	Q(t+1)	T
0	0	0
0	1	1
1	0	1
1	1	0

Design with JK-Flip Flop

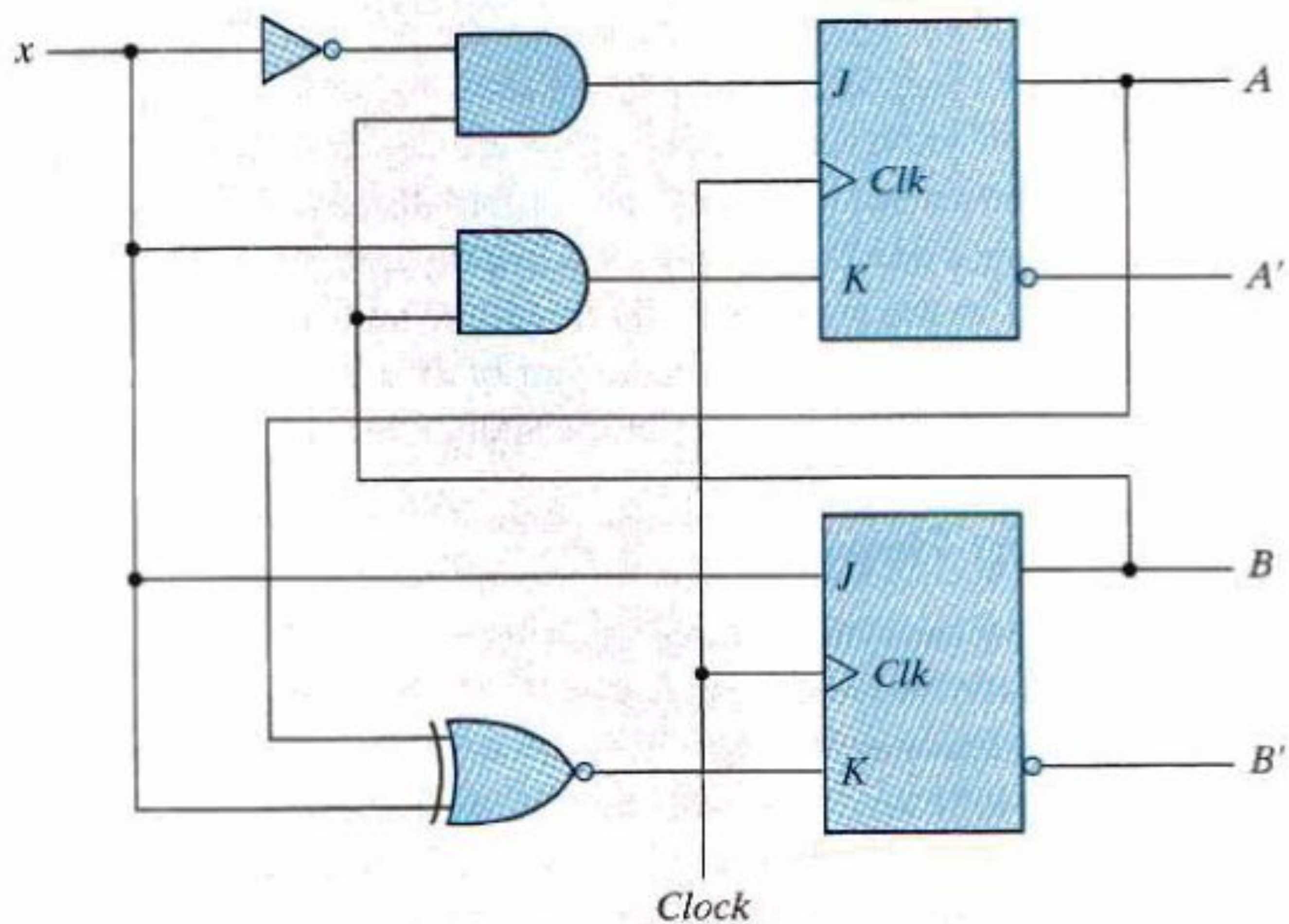
- For a given description, say the following state-table is obtained. The circuit has to be designed using JK Flip-Flops

Present State		Input	Next State		Flip Flops Inputs			
A	B	x	A	B	J_A	K_A	J_B	K_B
0	0	0	0	0	0	X	0	X
0	0	1	0	1	0	X	1	X
0	1	0	1	0	1	X	X	1
0	1	1	0	1	0	X	X	0
1	0	0	1	0	X	0	0	X
1	0	1	1	1	X	0	1	X
1	1	0	1	1	X	0	X	0
1	1	1	0	0	X	1	X	1

Design with JK-Flip Flop



Design with JK flip flop



Design with T-Flip Flop